



Mechanical Unit Operations (CHL2050)

Course Instructor

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Chapter 2: Filtration and Membrane Separation

1. theory
2. filtration equipment
3. membrane processes
4. general equation
5. microfiltration
6. fouling
7. pervaporation

Filtration

Fundamental Principle:

- Fluid flows through a filter medium due to a **pressure differential** across the medium
- Upstream pressure > Downstream pressure

Sources of Pressure Differential

- **Pressure above atmospheric (upstream side):**
 - Gravity (column of liquid)
 - Pump or blower
 - Centrifugal force
- **Vacuum (downstream side):**
 - Atmospheric pressure upstream, vacuum downstream

Gravity Filters – A Special Case

Limitations:

- Filter medium can be no finer than a **coarse screen** or a **bed of coarse particles** (e.g., sand)
- **Industrial applications (restricted):**
 - Draining liquor from **very coarse crystals**
 - **Clarification of potable water**
 - **Wastewater treatment**

Three Categories

1. **Pressure filters** – Pressure applied upstream
2. **Vacuum filters** – Vacuum applied downstream
3. **Centrifugal separators** – Centrifugal force drives filtration

Continuous vs. Discontinuous (Batch) Filters

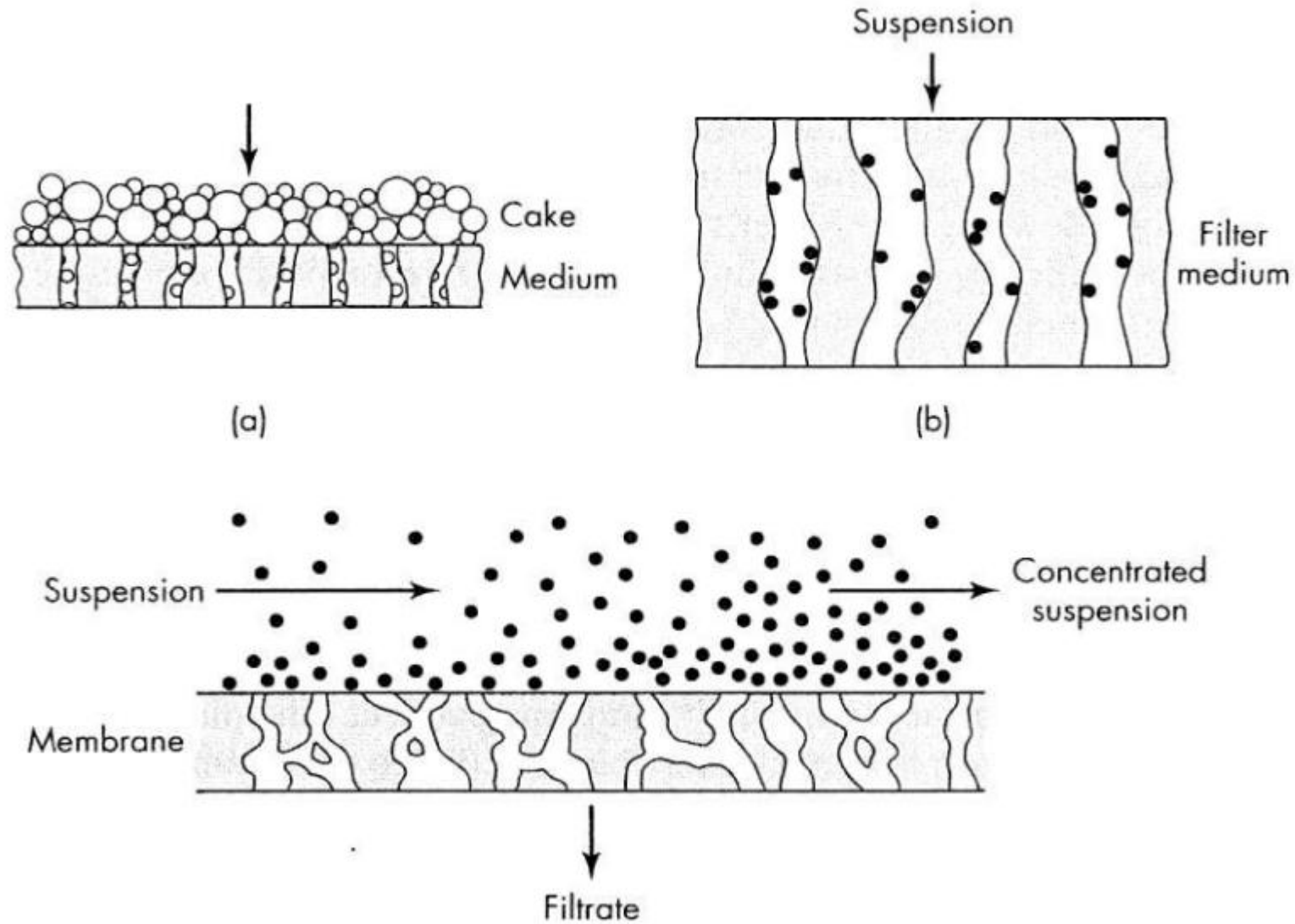
1. Discontinuous (Batch):

- Flow is continuous during operation
- Must be **interrupted periodically** to discharge accumulated solids

2. Continuous:

- Discharge of **both solids and fluid** is uninterrupted
- Equipment runs steadily without stopping

Filtration



Filtration

1. Cake Filters

- Separate **relatively large amounts of solids** as a cake (crystals or sludge)
- **Key features:** Often include provisions for: Washing the cake and removing some liquid from solids before discharge

2. Clarifying Filters

- Remove **small amounts of solids** to produce: Clean gas, Sparkling clear liquids (e.g., beverages)
- **How they work:**
- Solid particles are trapped **inside** the filter medium or on its **external surfaces**

Critical distinction from screens:

- Pores of the filter medium are **much larger in diameter** than the particles to be removed
- Particles are captured by **depth filtration or surface attachment**, not simple size exclusion

3. Crossflow Filters

- Feed suspension flows under pressure at **fairly high velocity across** the filter medium
- **How they prevent cake buildup:**
- Thin layer of solids may form on the surface
- **High liquid velocity** keeps the layer from building up

Filter medium:

- Ceramic, metal, or **polymer membrane**
- Pores small enough to exclude most suspended particles

Flow pattern:

- Some liquid passes through medium as **clear filtrate**
- Remaining suspension becomes **more concentrated**

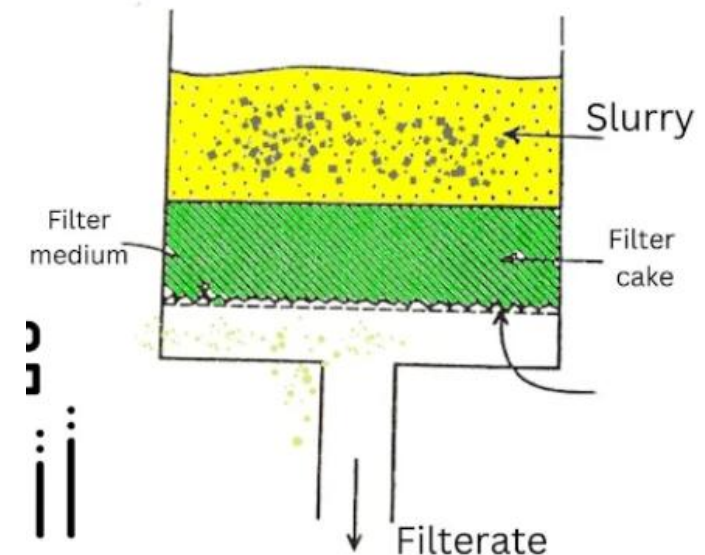
Cake filter

How a Cake Filter Works

- At the beginning of filtration, some solid particles enter and become trapped within the pores of the filter medium.
- Very quickly, additional particles accumulate on the surface of the medium.
- A visible layer of solids called the **filter cake** develops.
- After this initial period, the **cake itself becomes the main filtering medium**, not the original filter cloth or septum.

Important Features

- Cake thickness increases with filtration time.
- The cake must be removed periodically.
- Used primarily for **solid-liquid separations**.
- Filtration can be driven by:
 - Positive pressure (pressure filters)
 - Vacuum (vacuum filters)
- Can operate in either:
 - Batch (discontinuous) mode
 - Continuous mode



Filter Press (Discontinuous)

Main Components:

- **Set of plates** (square/circular, vertical/horizontal)
- **Filter medium** (canvas or polypropylene cloth)
- **Frames** (in plate-and-frame design)

Arrangement:

- Plates and frames sit **vertically** in a metal rack
- Cloth covers each plate face
- Assembly squeezed tight by **screw or hydraulic ram**

How a Filter Press Works

- **Slurry entry**
 - Enters at one end of plate/frame assembly
 - Passes through **lengthwise channel** (corner of assembly)
 - Auxiliary channels carry slurry into **each frame**
- **Filtration inside each frame**
 - Solids deposit on **cloth-covered plate faces**
 - Liquid passes through cloth
 - Filtrate flows down **grooves/corrugations** in plate faces
 - Exits the press

Types of Filter Press Construction

1. Recessed Plate Filter Press

- Chambers formed by recesses in molded plates.
- Most common modern design.
- Simpler operation.

2. Plate-and-Frame Filter Press

- Alternating plates and open frames.
- Solids accumulate inside frames.
- Classical design still found in many industries.

Why Wash the Cake?

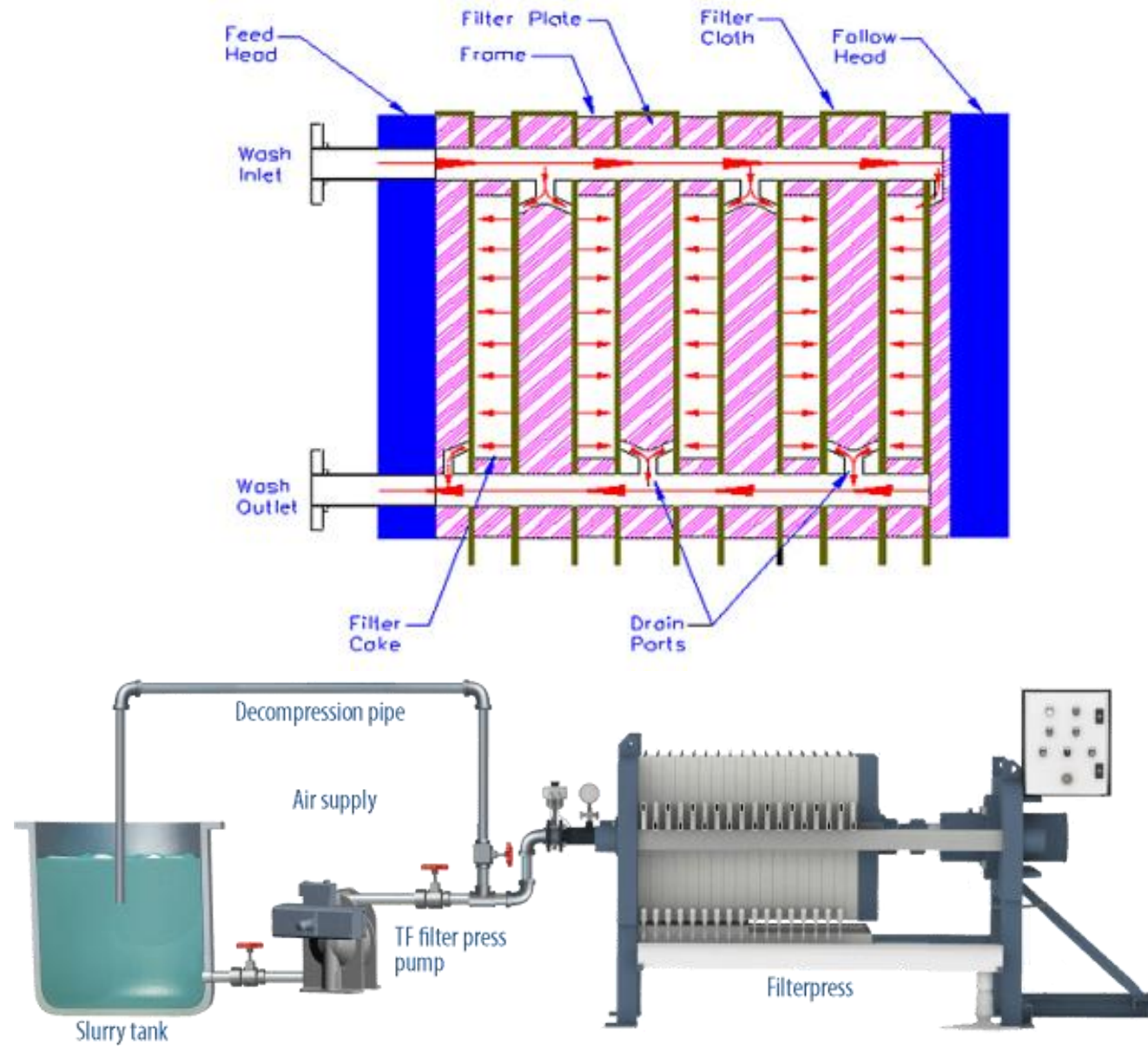
- To remove: Soluble impurities , Mother liquor
Contaminants

Cake Drying: After Washing Residual liquid remains trapped inside cake pores.

- **Drying Method:** Compressed air, Steam blowing
- **Objective:** Reduce moisture content before cake discharge.

Cake Discharge: Press is opened, Plates separated, Solid cake falls out by gravity.

Filter Press



Rotary Vacuum Filter

- **Most common type** of continuous vacuum filter
Vacuum applied from a **stationary source** to moving parts via a **rotary valve**

Main Components:

- **Horizontal drum** with slotted face
- **Rotates at:** 0.1 to 2 r/min (very slow)
- **Partially submerged** in an agitated slurry trough

Inside the Drum Under the slotted outer drum → a **smaller inner drum** with solid surface

- Between them: **Radial partitions** dividing annular space into **separate compartments**
- **Filter Medium:** Canvas or cloth covering the drum face
- **Rotary Valve:** Each compartment connects via internal pipe to one hole in the rotating valve plate
- **Key point:** Vacuum and air are **alternately applied** to each compartment as the drum rotates

Filtration Cycle

Stage 1: Cake Formation (Submerged zone)

- Panel enters slurry trough
- Vacuum applied through rotary valve
- Liquid drawn through cloth → into compartment → internal pipe → valve → collecting tank
- **Solids build up** as a layer on the panel face

Stage 2: Washing & Drying (Above slurry)

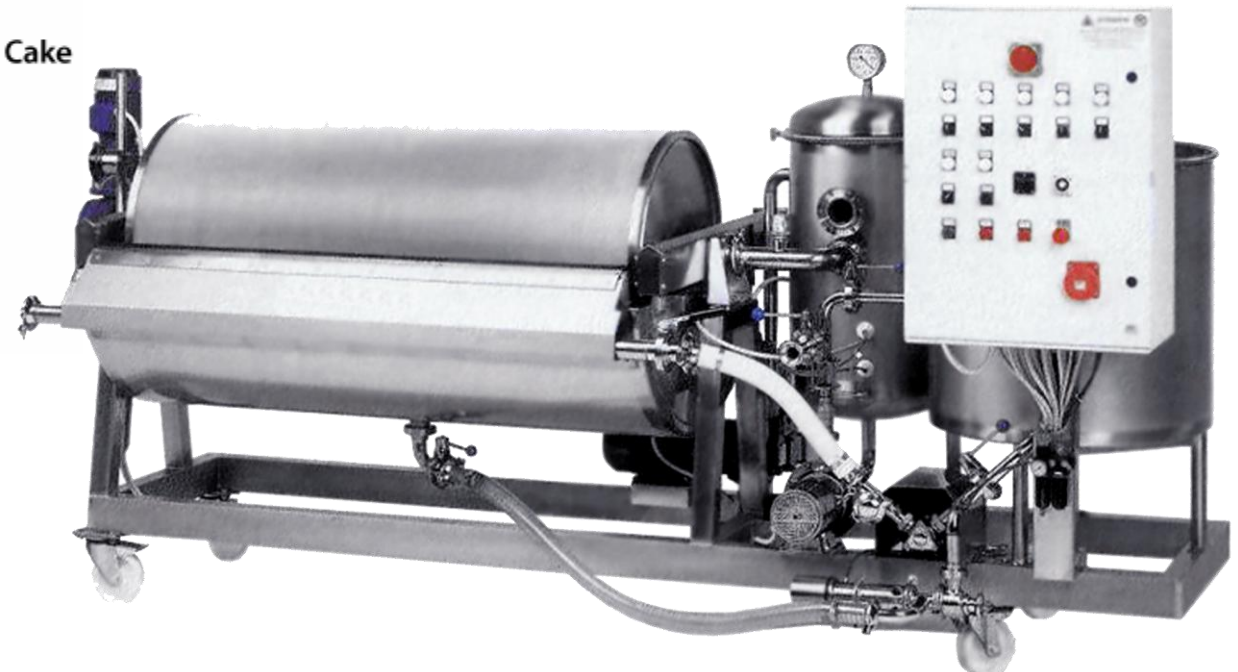
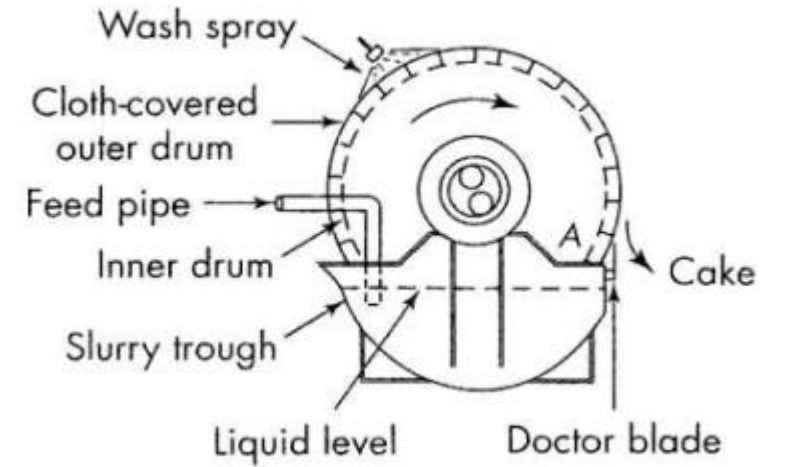
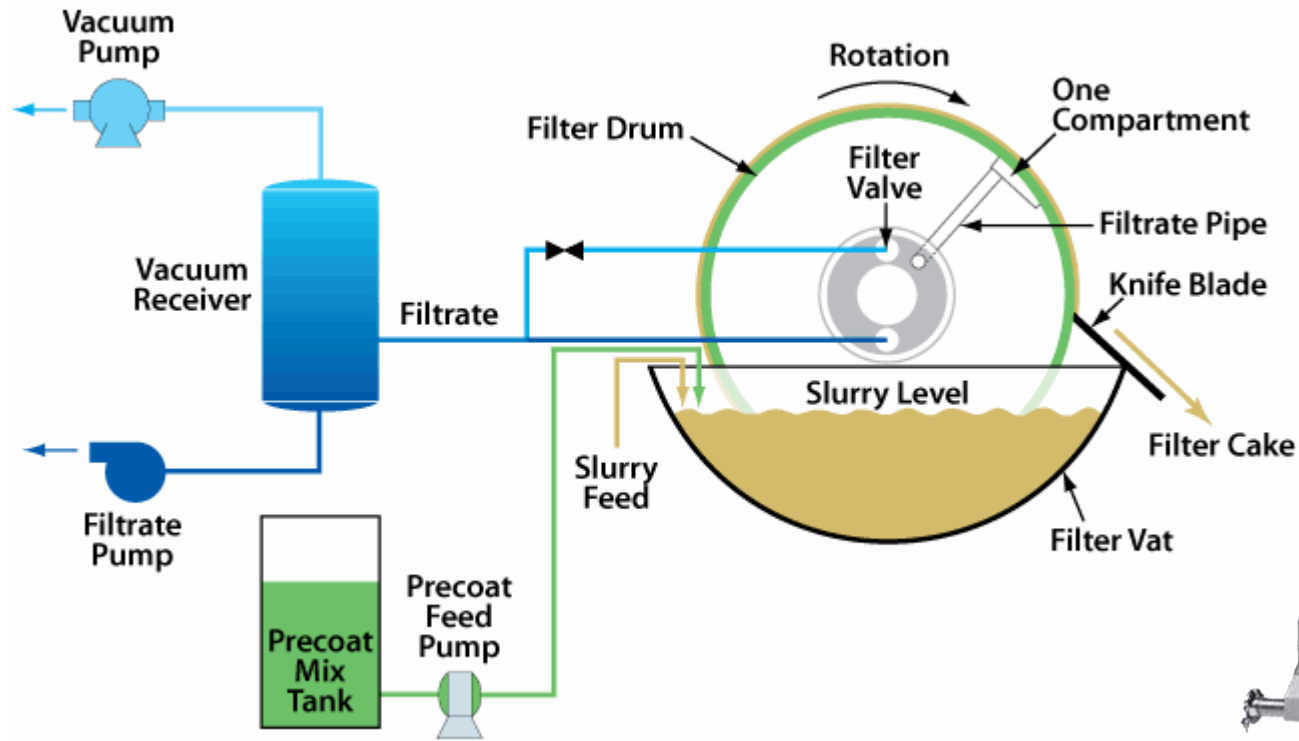
- Panel leaves slurry
- Vacuum applied from a **separate system**
- Wash liquid + air drawn through the cake

Stage 3: Cake Discharge (End of cycle)

- Panel leaves drying zone
- Vacuum cut off
- Cake removed by **doctor blade** (horizontal knife)
- **Small air blow** under cake → bellies out the cloth → cracks cake away
- *Blade does not touch the drum face*

Stage 4: Return Panel reenters slurry- Cycle repeats

Rotary Vacuum Filter



Rotary Vacuum Filter

Why It's Continuous? Per panel: Cyclic (formation → wash → dry → discharge → return)

Per filter: Continuous → Because different panels are in **different stages of the cycle at all times**

Fluid Paths (Two separate vacuum receivers):

Air Path:

- Dry vacuum pump pulls air out
- Moisture trap protects vacuum pump

Cake Path:

- Formed on drum → scraped by doctor blade → discharged
- **Wash Sprays:** Shown on top of cake (optional)
- **Barometric seal:** Prevents liquid from being sucked into vacuum pump

Stream	Destination
Filtrate (from cake formation)	Filtrate receiver → pump
Wash liquid (from washing zone)	Separate wash receiver → pump

Horizontal Belt Filter

Problem with rotary drum filters:

Coarse, fast-settling particles **cannot be kept suspended** in the slurry trough

Cake often **will not adhere** to the drum surface

Solution: Top-fed horizontal belt filter

Basic Design

Resembles: A belt conveyor

Key components:

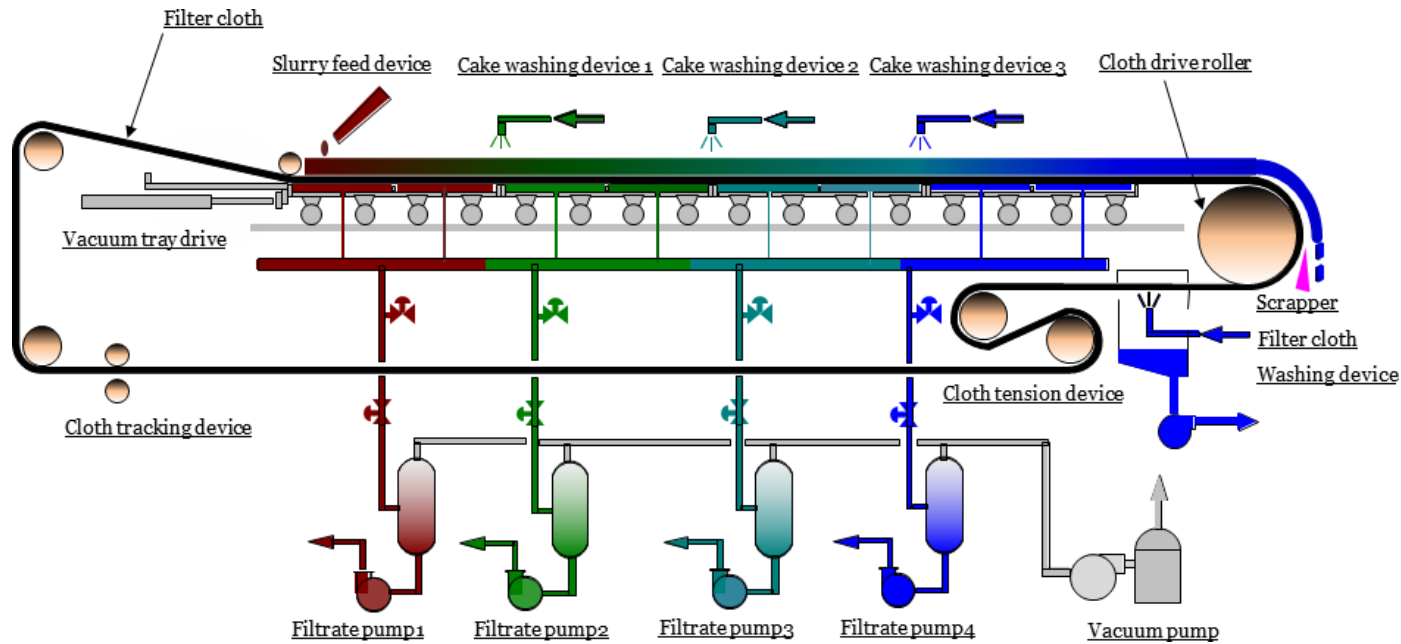
- **Drainage belt** (transversely ridged or supported) – endless loop
- **Filter cloth** – also an endless belt, carried by drainage belt
- **Longitudinal vacuum box** under the belt
- Central openings in drainage belt slide **over** the vacuum box

How filtrate is collected:

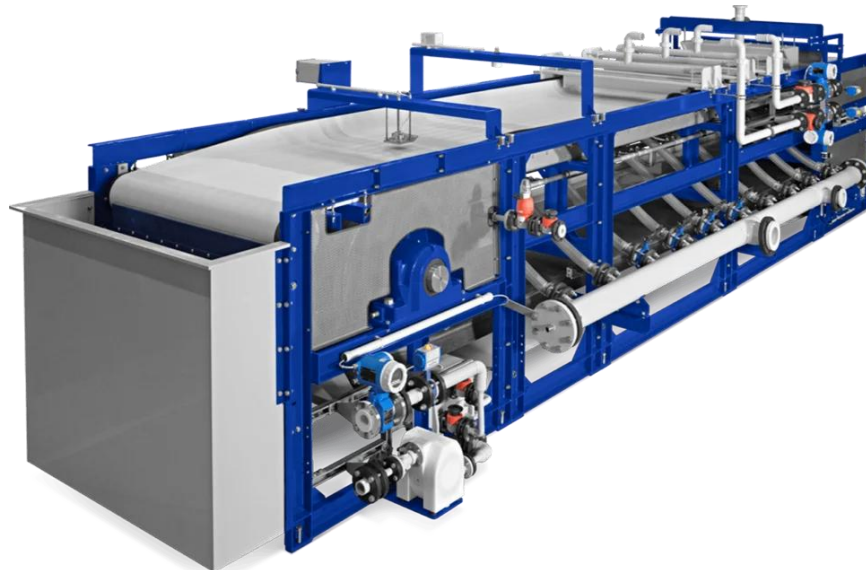
Filtrate is drawn **through cloth** → **through drainage belt openings** → **into vacuum box**

Applications: Municipal wastewater, Mining tailings, Power plant, Chemical plant

Horizontal Belt Filter



Step	Action
1	Feed slurry flows from distributor at one end onto belt
2	Vacuum pulls liquid through cloth
3	Cake forms on belt surface
4	Belt moves forward
5	Washed cake discharges from opposite end



Top-Suspended Batch Centrifugal filter

- Widely used industrial batch centrifuge for solid–liquid separation.
- Consists of a perforated basket suspended from the top.
- Typical basket diameters: **750–1200 mm**.
- Operates by centrifugal force to separate solids from liquids.

Main Components

- Motor drive
- Perforated rotating basket
- Filter medium/cloth lining basket wall
- Feed inlet
- Wash liquid inlet
- Solids cake zone
- Casing (housing)
- Liquid discharge outlet
- Adjustable knife
- Removable bottom plate

Filter Medium

- Mounted on the inner wall of the perforated basket.
- Acts as support for the solids cake.
- Allows liquid to pass while retaining solid particles.

Industrial Applications: Salt crystals, Drug intermediates, Fine chemicals, Sugar crystals, Crystalline and granular solids

Top-Suspended Batch Centrifugal filter

Working Principle

Step 1: Feeding

- Suspension is introduced through the feed inlet.
- Feed enters the rotating basket.

Step 2: Cake Formation

- Centrifugal force drives liquid outward through:
 - Filter medium
 - Perforated basket wall
- Solids are retained on the filter medium forming a cake.

Step 3: Filtrate Removal

- Liquid exits through the casing and leaves via the discharge pipe.

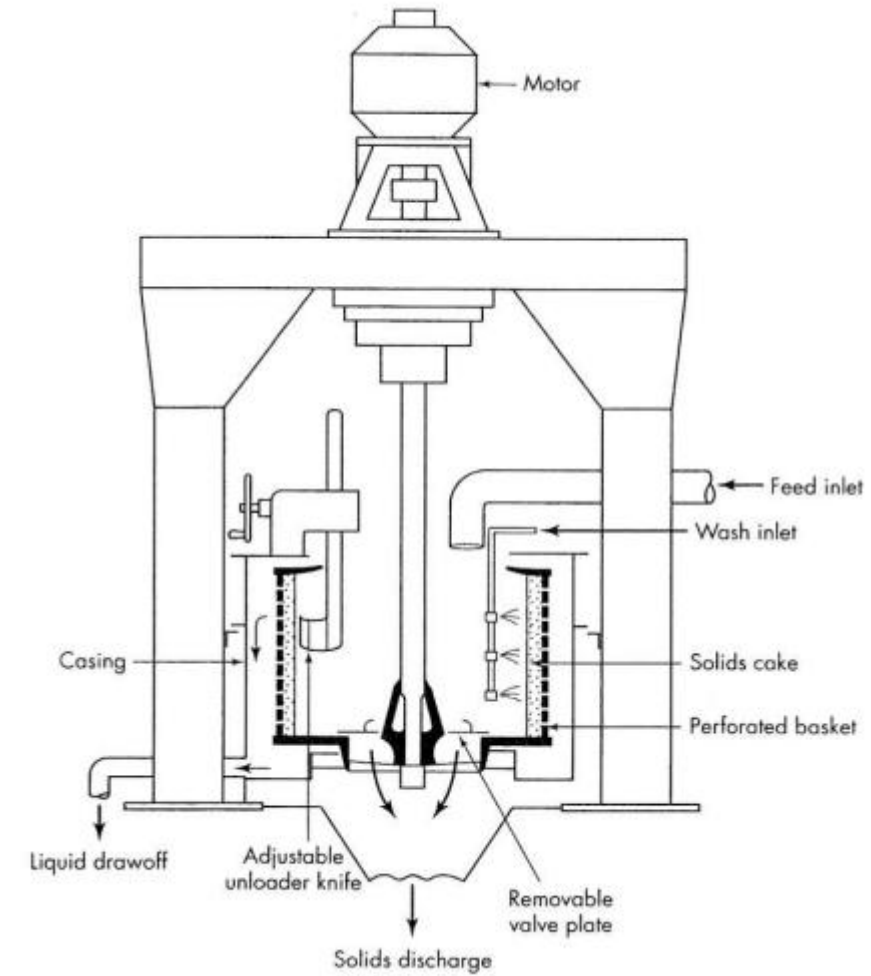
Batch Operating Sequence

1. Feed suspension into basket
2. Form solids cake
3. Drain filtrate
4. Wash cake (if required)
5. Dewater cake
6. Slow basket rotation
7. Scrape and discharge solids
8. Clean filter medium
9. Repeat cycle

Top-Suspended Batch Centrifugal filter

Advantages

- High centrifugal force.
- Efficient solid–liquid separation.
- Excellent cake washing capability.
- Produces relatively dry solids.
- Reduced vibration due to suspended design.
- Suitable for crystalline products.
- Can be partially or fully automated.



Filter Media

1. Filter Media: The porous barrier that retains solids while allowing fluid to pass.

2. Key Requirements for a Filter Medium:

- **Retention:** Must capture solids and produce a clear filtrate.
- **Anti-Plugging:** Must not blind or clog easily.
- **Chemical & Physical Resistance:** Withstand process temperature, pressure, and corrosive fluids.
- **Cake Discharge:** Allow easy and complete removal of the filter cake.
- **Cost-Effectiveness:** Must be economically viable for the application.

3. Common Materials for Filter Media:

- **Canvas Cloth:** Duck or twill weave (most common for general use).
- **Synthetic Fabrics:** Nylon, Polypropylene, Polyesters (high chemical resistance).
- **Metal Cloth:** Monel, Stainless Steel (for corrosive or high-temperature services).
- **Specialty:** Woolen cloth, Glass cloth, Paper.

4. Important Practical Note:

- Smooth synthetic/metal fibers are *less effective* than rough natural fibers for removing **very fine particles** initially.
- **However**, this is usually temporary. **Why?** The *actual* filtering medium quickly becomes the **first layer of deposited solids** (the initial cake), not the cloth itself.
- **Result:** Filtrate may start *cloudy* but becomes *clear* as the cake builds. Cloudy initial filtrate is recycled back to the feed tank.

Filter aids

The Problem:

- Slimy or very fine solids form a **dense, impermeable cake** that quickly **plugs** the filter medium.
- Result: Filtration rate drops drastically.

2. The Solution – Filter Aids:

- Add an inert, porous material to the slurry *before* filtration.
- **Purpose:** Increase the **porosity** of the cake → Allows liquor to pass at a reasonable rate.

3. Common Filter Aid Materials:

- Diatomaceous Silica (Diatomaceous Earth)
- Perlite
- Purified Wood Cellulose
- Other inert porous solids

Two main ways to use filter aids

Method	Description	Application
Body Feed (Slurry Addition)	Mixed directly with the feed slurry before filtration.	When solids have no value – discarded together with filter aid.
Precoating	A layer of filter aid is deposited on the medium before filtration begins.	Prevents gelatinous solids from plugging the medium; gives clearer filtrate.

Cake Filtration – Fundamentals & Flow Resistance

1. Filtration as Flow Through Porous Media:

- Filtration is a special case of flow through porous media.
- **Key Difference:** In filtration, flow resistances are **not constant** – they **increase with time**.

2. Why Does Resistance Increase?

- Filter medium becomes clogged.
- Filter cake builds up on the medium.
- **Result:** Either flow rate decreases (if pressure is constant) OR pressure drop increases (if flow rate is constant).

Two possible operating scenarios

Mode	Pressure Drop (ΔP)	Flow Rate (Q)	Typical Use
Constant-Pressure Filtration	Held constant	Decreases with time	Most common in industry
Constant-Rate Filtration	Progressively increased	Held constant	Less common; used when constant throughput is required

Two Resistances in Series:

- Liquid passes through **two resistances** arranged in series:
- **Cake Resistance** – Due to the filter cake.
- **Medium Resistance** – Due to the filter cloth/medium.

Resistance Behavior Over Time:

- **Medium Resistance:** Important only during **early stages**; remains relatively constant.
- **Cake Resistance:** **Zero at start**; **increases with time** as cake builds up.
- **During Washing:** Both resistances are constant; medium resistance is usually negligible.

Cake Filtration: Pressure Drop Distribution

1. Overall Pressure Drop: The total pressure drop across the entire filter is the **sum** of the individual pressure drops across the cake and the medium.

2. Pressure Drop Components:

p_a = Inlet pressure (upstream of cake)

p' = Pressure at the boundary between cake and medium

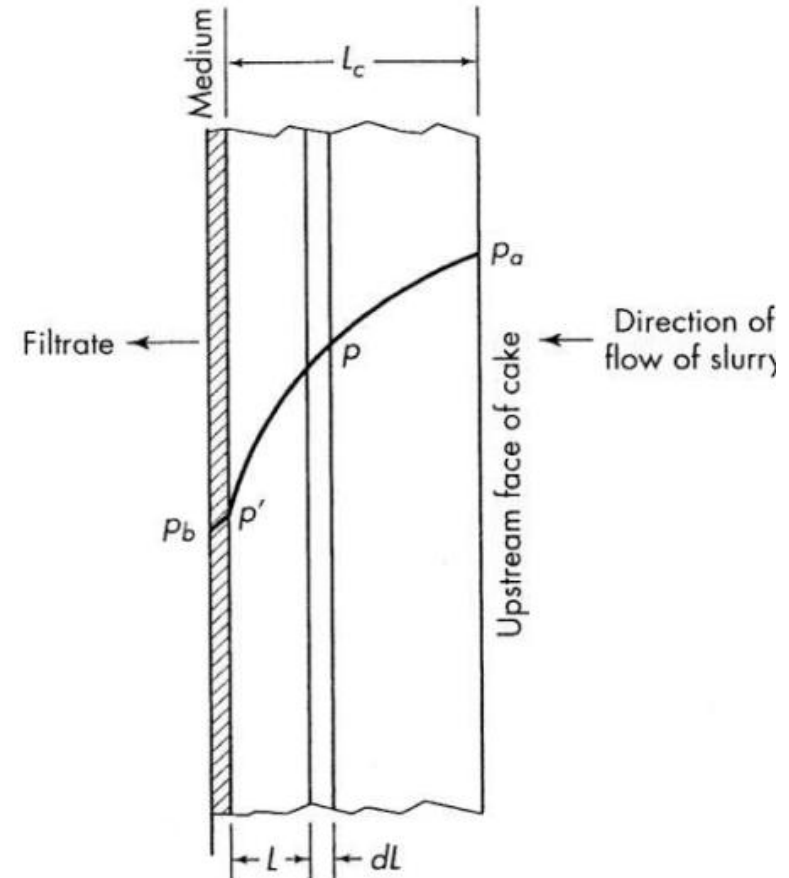
p_b = Outlet pressure (downstream of medium)

3. Mathematical Representation:

- Overall $\Delta p = \Delta p_c + \Delta p_m$

Where:

- $\Delta p_c = (p_a - p') =$ Pressure drop across the cake
- $\Delta p_m = (p' - p_b) =$ Pressure drop across the medium
- $\Delta p = (p_a - p_b) =$ Total pressure drop



Clarifying Filters

1. Clarifying Filters: Filters designed to remove **small amounts** of solids or liquid droplets from liquids or gases. Used for "**polishing**" – removing trace contaminants to achieve high clarity.

2. Key Distinction from Screening:

- In **screening**: Pores are **smaller** than particles → particles are *mechanically trapped* by size exclusion.
- In **clarification**: Pores are **larger** – sometimes *much larger* – than the particles to be removed.

So how do they work?

- Particles are captured by **surface forces** (e.g., electrostatic, van der Waals, adsorption).
- Particles immobilize on surfaces or within flow channels.
- They **reduce the effective diameter** of channels but typically **do not block them completely**.

3. Typical Feed Characteristics:

- Very dilute suspensions – generally **no more than 0.10% solids** by weight.
- Examples: Beverages, pharmaceuticals, lubricating oils, electroplating solutions.

4. Applications:

- **Polishing** beverages, pharmaceutical products, fuel oil, lubricants.
- **Essential** for cleaning feed in:
 - Fiber spinning processes.
 - Film extrusion processes.
- Also used in water treatment (gravity-bed filters).

1. Equipment Types for Liquid Clarification:

- **Gravity-bed filters** – for water treatment (coarse media).
- **Disk and plate presses** – for fine clarification.
- **Cartridge clarifiers** – disposable or cleanable elements.
- **Tank filters** – used extensively.
- **Continuous precoat filters** – for demanding applications.
- **Crossflow filters** – covered in the next section.

2. Example: Disk Clarifying Filter

Construction: Pack of disks made of **asbestos and cellulose fibers**.

Operation:

- Assembly sealed into a **pressure case**.
- Typical gauge pressure: **less than 345 kPa (50 psi)**.
- Liquid flows **inward** through the disks.
- Filtrate collected into a **central manifold**.
- Discharge: Central or peripheral.

Capacity: Up to **378 L/min (6,000 gal/h)** for low-viscosity liquids.

3. Operating Characteristics:

Batch Operation:

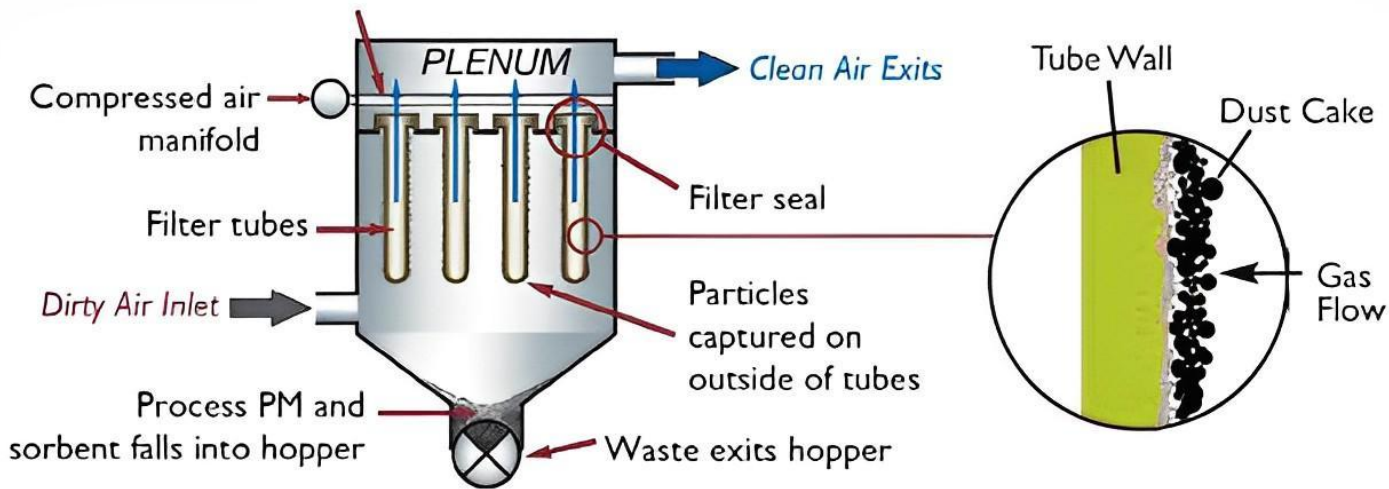
- Filtration rate and solids removal efficiency are **constant** for a considerable period.
- Eventually, filtrate solids content rises to an unacceptable "**breakthrough**" value.
- **Action Required:** Backwashing of the filter element becomes necessary to restore performance.

Types of Gas Cleaning Filters:

Why Gas Cleaning?: Remove dust and particulate matter from air and industrial gases. Essential for:

- Environmental compliance (emission control).
- Product quality (e.g., food, pharmaceutical manufacturing).
- Equipment protection (e.g., gas turbines, compressors).

Filter Type	Description	Application
Pad Filters	Pads of cellulose pulp, cotton, felt, glass fiber, or metal screening	Atmospheric dust; light-duty air cleaning
Granular Bed Filters	Stationary or moving beds of granules (30-mesh to 8-mesh, or 12–40 mm)	Process dusts; heavy-duty applications
Bag Filters	Large bags of felt or woven fabric in a metal housing	Process dusts; high-efficiency (99%+)



Pad Filters

- **Media:** Cellulose pulp, cotton, felt, glass fiber, or metal screening.
- **Enhancement:** May be **dry** or coated with **viscous oil** (acts as dust holder).
- **Disposable vs. Reusable:**
 - **Light duty:** Pads are **non-recyclable**.
 - **Large-scale:** Pads are **rinsed and recoated** with oil.

Granular Bed Filters

- **Media:** Stationary or moving beds of granules.
- **Granule Size Range:**
 - Fine: **30-mesh to 8-mesh**.
 - Coarse: **12 to 40 mm** ($\frac{1}{2}$ to $\frac{1}{4}$ in.).
- **Mechanism:** Particulates are trapped within the bed as gas flows through.

Bag Filters – Construction:

One or more **large bags** of **felt** or **thin woven fabric**.

Mounted inside a **metal housing** (baghouse).

Flow Configuration:

Typical: Dust-laden gas enters at the **bottom** and passes **outward** through the bag, leaving dust behind on the inside surface.

Alternative: Flow can be **inward** in some designs.

Cleaning Cycle (Cake Discharge):

Periodically, flow is automatically **cut off**.

Dust is dislodged by:

Back-blowing: Clean gas is blown in the reverse direction.

Mechanical shaking: Bags are shaken to release the dust.

Dislodged dust is collected for **recovery** or **disposal**.

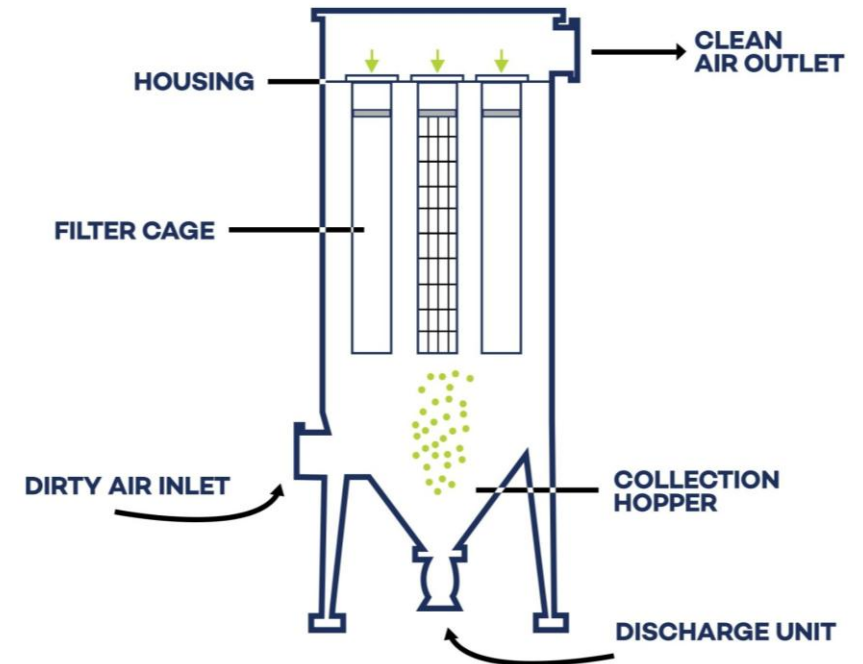
Separation Mechanism:

In all these gas cleaning filters, most of the separation is by **impingement**.

Impingement: Particles collide with the filter medium or granules and are captured due to inertia, interception, or diffusion.

Filtration

Bag Filters



Membrane separation

"Difficult" Materials: Certain substances are inherently expensive and challenging to separate.

Category A: Finely Dispersed Solids

- Compressible cakes (filtration issues).
- Density close to the liquid phase (slow settling).
- High viscosity or gelatinous nature (poor flow/filtration).

Category B: Molecular & Ionic Solutes

- Low molecular weight, non-volatile organics (e.g., pharmaceuticals).
- Dissolved salts (requiring energy-intensive thermal methods).

Category C: Biological Materials

- Highly sensitive to physical/chemical environment (heat, pH, shear).
- **Example:** Proteins, enzymes, and cellular components.

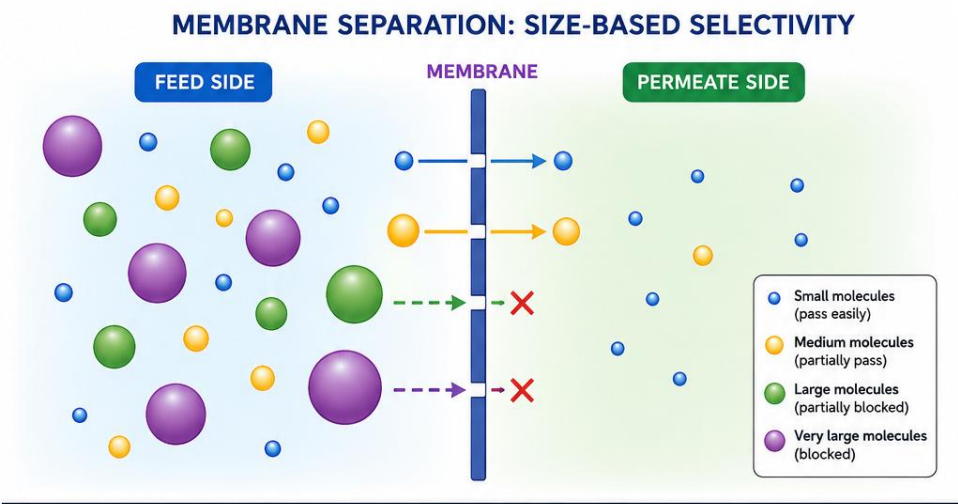
What is a Membrane? *An interphase separating two phases and selectively controlling the transport of materials between those phases.*

Key Concept: "Interphase" vs. "Interface": A membrane is a *finite, three-dimensional* barrier, not just a two-dimensional boundary.

Key Advantages over Conventional Processes (e.g., Distillation, Filtration):

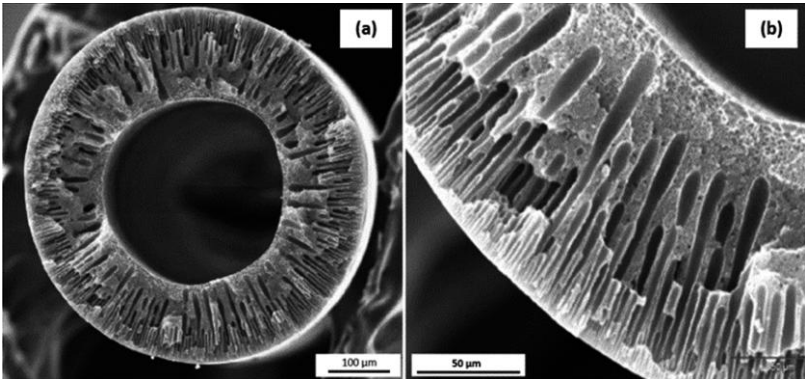
- ✓ **Ambient Temperature Operation:** Preserves heat-sensitive materials.
- ✓ **Relatively Low Capital & Running Costs:** Lower energy consumption (no phase change).
- ✓ **Modular Construction:** Scalable and easy to expand.

Classification of Membranes



Name of Process	Driving Force	Separation Size Range	Examples of Materials Separated
Microfiltration (MF)	Pressure Gradient	10 – 0.1 μm	Small particles, large colloids, microbial cells
Ultrafiltration (UF)	Pressure Gradient	<0.1 μm – 5 nm	Emulsions, colloids, macromolecules, proteins
Nanofiltration (NF)	Pressure Gradient	$\sim$ 1 nm	Dissolved salts, small organics
Reverse Osmosis (RO)	Pressure Gradient	<1 nm	Dissolved salts, small organics (desalination)
Electrodialysis	Electric Field Gradient	<5 nm	Dissolved salts
Dialysis	Concentration Gradient	<5 nm	Treatment of renal failure (hemodialysis)

Classification of Membranes



Microfiltration (MF)	Ultrafiltration (UF)	Nanofiltration (NF)	Reverse Osmosis (RO)
• Sterilization of beverages	• Paint recovery in auto plants	• Pharmaceutical purification	• Desalination of seawater
• Clarification of juices	• Dairy (whey protein conc.)	• Water softening (removal of hardness ions)	• Dairy (whey concentration)
• Filtration in wastewater treatment	• Industrial Savings: 10-30% reduction in paint usage.	• Dye and sugar concentration	• Production of ultrapure water
• Cell harvesting in biotech			• Concentration of fruit juices

Nature of Synthetic Membranes

1. What are Synthetic Membranes?

- Engineered barriers used in pressure-driven processes (MF, UF, RO) and dialysis.
- They selectively allow the passage of certain components (permeate) while retaining others (retentate).

2. Dominant Material: Polymers

- Most common type for industrial membrane separations.
- **Early Generation:** Cellulosic materials (e.g., cellulose acetate).
- **Modern Advancements:** Synthetic polymers (Polyamide, Polysulphone, Polycarbonate).
- **Key Benefits:** Improved chemical stability and superior resistance to microbial degradation.

3. Membrane Formats

- Membranes are manufactured in various physical forms to suit different module designs:
- **Flat Sheets:** Most common form.
- **Tubular:** Diameter 10-25 mm.
- **Hollow Fibres:** Diameter 0.1-2.0 mm.

Manufacturing Polymeric Membranes: Phase Inversion

The "Immersion Precipitation" Technique

- The most common method for creating polymeric membranes.
- Based on the principle of **phase separation** to create a porous structure.

The 4-Step Process:

1. **Dissolution:** Polymer (10-30% by mass) is dissolved in a suitable solvent.
2. **Casting:** The polymer solution (or "dope") is spread onto a support as a thin film (~100 μm thick)
3. **Quenching (Immersion):**
 - The film is immersed in a **non-solvent bath** (usually water).
 - **Mass Transfer:** Solvent leaves the film, non-solvent enters.
 - **Phase Separation:** The solution separates into two phases:
 - **Polymer-rich phase** -> Forms the solid membrane matrix.
 - **Polymer-depleted phase** -> Forms the pores.
4. **Annealing:** The membrane is heated to stabilize the final structure and optimize its properties.

1. Inorganic Membranes

Materials: Zirconia, Alumina, Titania.

Production Techniques:

- Deposition of colloidal metal oxides on a support.
- High-temperature sintering of ceramic microspheres.

Advantages over Polymers:

- Higher temperature stability (enables steam sterilisation).
- Increased resistance to fouling (clogging).
- Narrower pore size distribution.

2. Membrane Characterisation

Direct Microscopy:

- **Electron Microscopy:** Can resolve pores down to ~10 nm.
- **Atomic Force Microscopy (AFM):** Resolves pores <1 nm, provides 3D structure data

Indirect / Permeability Methods:

Bubble point, Mercury intrusion - used to determine maximum pore size and pore size distribution.

3. Nominal Molecular Weight Cut-Off (MWCO)

- **Definition:** The molecular weight of a solute for which the **rejection coefficient (R) = 0.95** (i.e., 95% of the solute is retained by the membrane). (*Refer to Figure 8.3*).
- **Concept:** It's an empirical characterization, not an absolute physical cut-off.
- **Typical Values:** For UF membranes, MWCOs range from 2,000 to 100,000 kg/kmol.

Membrane Structures: Symmetric vs. Asymmetric

1. Symmetric (Isotropic) Membranes

- **Structure:** Uniform pore structure and density across the entire membrane thickness.
- **Example:** Most **Microfiltration (MF)** membranes.

Key Feature: Can have a very high porosity (up to 80%), allowing for high flow rates but limited selectivity.

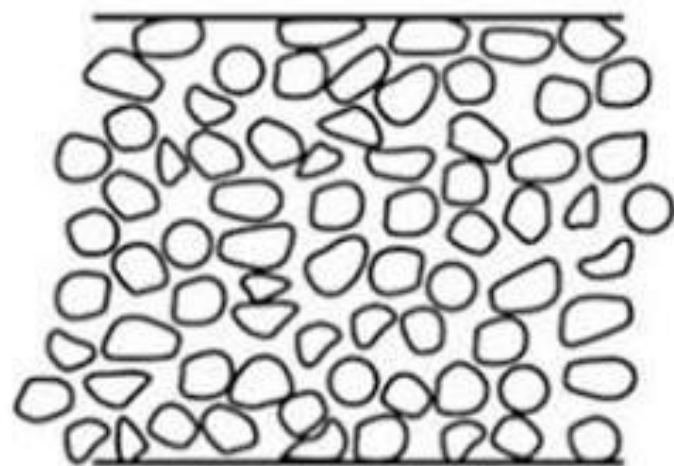
2. Asymmetric (Anisotropic) Membranes

Structure: Non-uniform across the thickness. It consists of two distinct layers (see Figure 8.1):

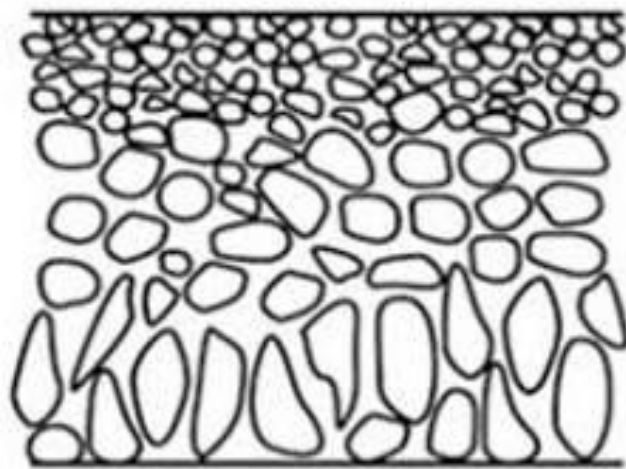
- **A Dense "Skin" Layer:** Very thin (1-2 μm). Contains the finest pores and is responsible for the membrane's selectivity.
- **A Porous Support Layer:** Much thicker ($\sim 100 \mu\text{m}$). Has large, open pores and provides mechanical strength with minimal flow resistance.

Example: **Ultrafiltration (UF)** and **Reverse Osmosis (RO)** membranes.

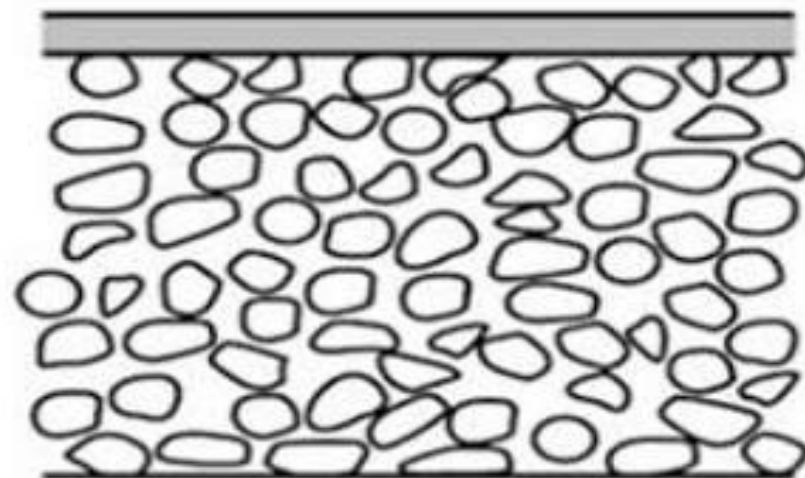
Key Feature: *Essential for high permeation rates.* The thin selective layer offers low resistance to flow, while the thick support prevents the membrane from collapsing.



(a) Symmetric



(b) Asymmetric



(c) Thin-film composite asymmetric membrane

Ion-Exchange Membranes

1. Ion-Exchange Membranes (IEMs):

- Charged, highly swollen polymeric gels used specifically in **electrodialysis** (and other electromembrane processes). They allow the selective transport of ions under the influence of an electric field.
- **Function:** They are not pressure-driven like RO/UF, but are driven by electrical potential.

2. Membrane Types & Mechanism

a) Cation-Exchange Membranes (CEMs):

- **Structure:** Contain fixed **negatively charged** groups (e.g., sulphonic acid, $-\text{SO}_3^-$, or carboxylic acid, $-\text{COO}^-$) attached to the polymer backbone.
- **Function:** These fixed negative charges **exclude (repel) anions** (co-ions) from entering the membrane. Therefore, they **selectively transport cations** (e.g., Na^+ , Ca^{2+}).

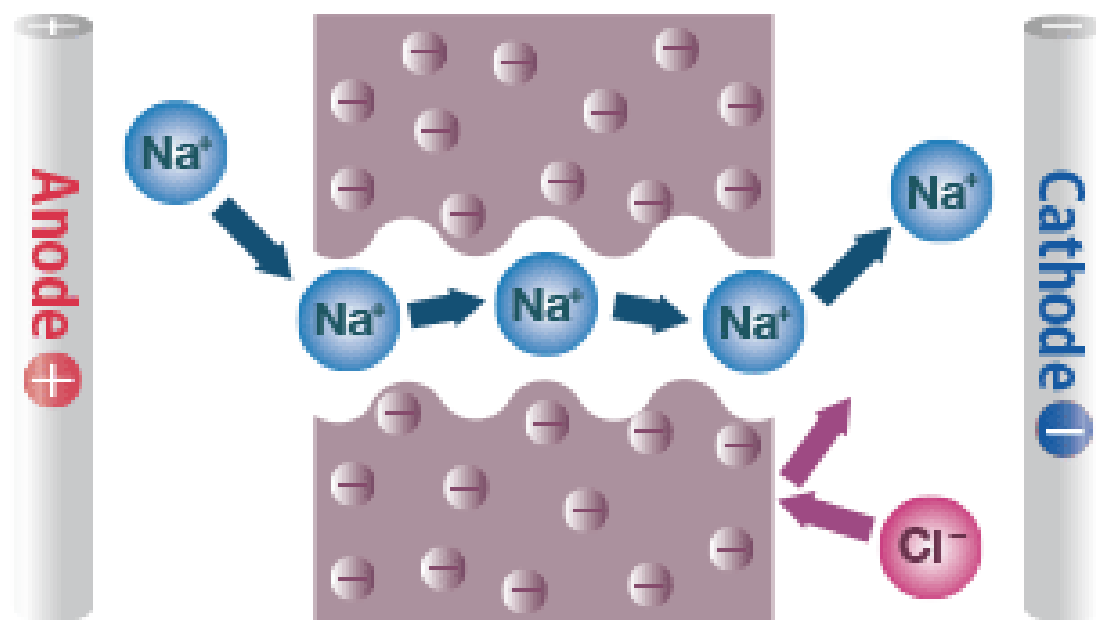
b) Anion-Exchange Membranes (AEMs):

- **Structure:** Contain fixed **positively charged** groups (e.g., quaternised amine groups, $-\text{NR}_3^+$) attached to the polymer backbone.
- **Function:** These fixed positive charges **exclude (repel) cations** (co-ions) from entering the membrane. Therefore, they **selectively transport anions** (e.g., Cl^- , NO_3^-).

3. Membrane Fabrication Types

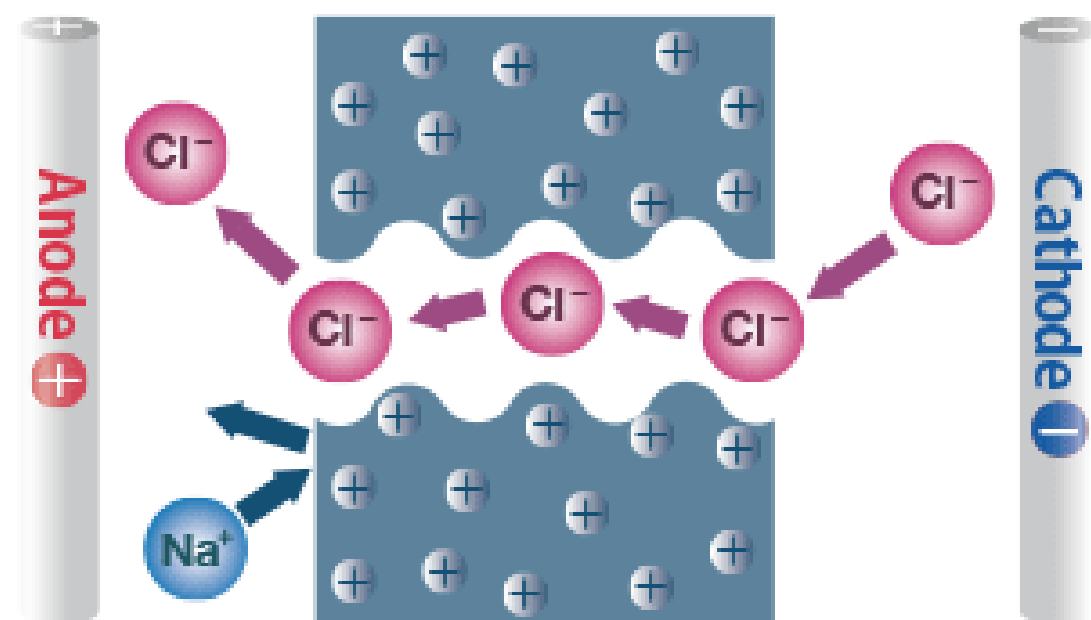
- **Heterogeneous Membrane:** Ion-exchange material (powder/resin) is dispersed within an inert polymer binder.
- **Homogeneous Membrane:** Charged groups are chemically bonded directly into the polymer matrix itself. (These are generally preferred for better electrochemical performance).

Membrane



cation exchange membrane

Figure 1-1 Selective permeability of CEM



anion exchange membrane

Figure 1-2 Selective permeability of AEM

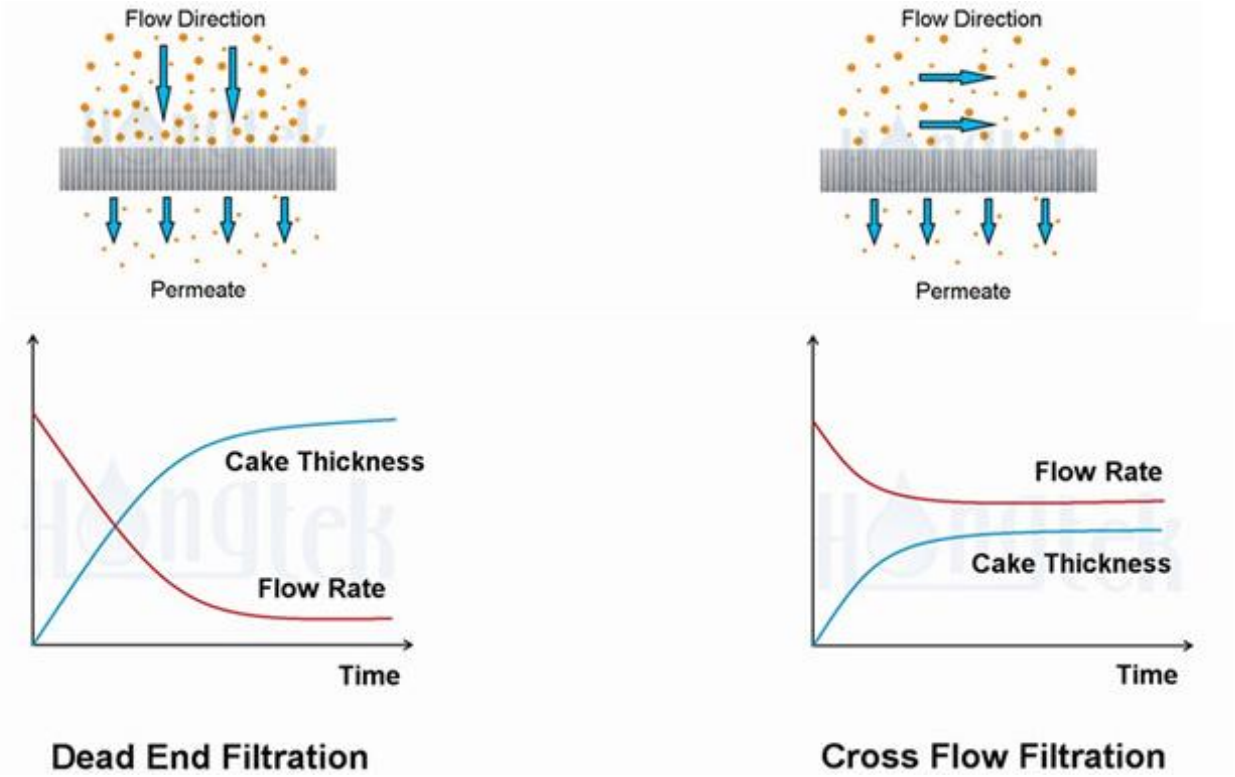
Challenge of Fine Particle Filtration - Sub-10 μm Problem:

- Conventional filtration (e.g., using filter cloths) becomes very difficult for particles smaller than about 5–10 μm .
- **Why?** The particles are so small that they rapidly form an impermeable, highly compressed filter cake. The resistance to flow (R_c) becomes enormous in a very short time.
- The filtrate flow rate drops to uneconomically low levels.

Conventional Approach: Slurry Thickeners

- One traditional solution is to use a slurry thickener (e.g., a gravity settler or centrifuge).
- Here, the goal is *not* to form a filter cake, but to discharge a concentrated slurry continuously.
- **Limitation:** These are generally limited to concentrating particles above $\sim 5 \mu\text{m}$. They struggle with very fine colloids and sub-micron particles.

Dead-End vs. Cross-Flow Filtration



Feature	Dead-End Filtration	Cross-Flow Filtration
Flow Direction	Perpendicular to membrane	Parallel to membrane
Cake Formation	Thick, rapidly growing cake	Thin, self-limiting cake
Flux Decline	Rapid and severe	Gradual and manageable
Suitability	Low solids, "clean" fluids	High solids, "dirty" feeds

Cross-Flow Microfiltration (CFMF)

The Concept:

- The particle-laden fluid is pumped at high velocity (1–8 m/s) across the face of a porous membrane.
- A pressure difference of 0.1–0.5 MPa (1–5 bar) is applied across the membrane.
- **Permeate:** The liquid that passes through the membrane (clean filtrate).
- **Retentate:** The concentrated slurry that exits the module.
- **The Module:** Can be rectangular (flat sheet) or tubular in geometry.

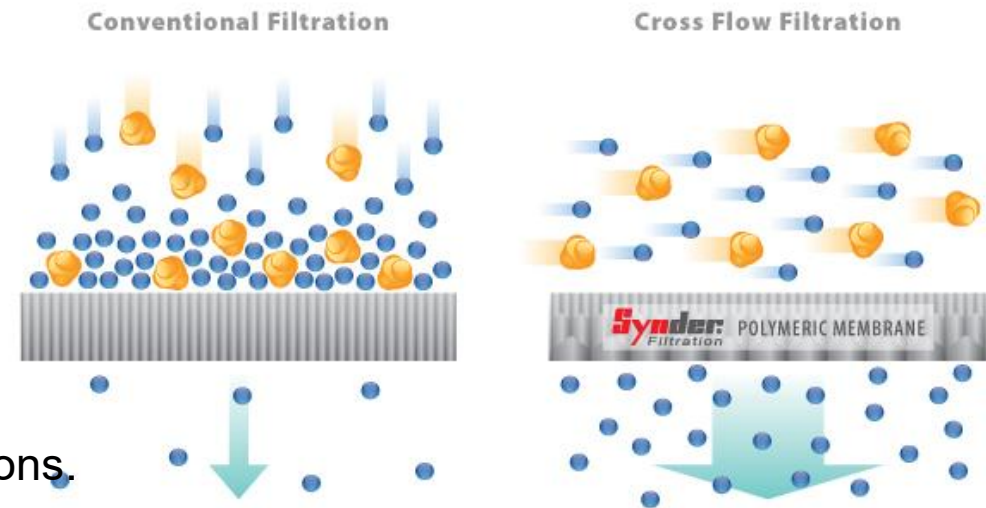
The Schematic (Figure 8.4):

- Feed enters the module with a high cross-flow velocity.
- Permeate passes through the membrane.
- Retentate (concentrated feed) exits at the other end.

Microfiltration Scope:

- Removes particles down to $<0.1\ \mu\text{m}$.
- Used for clarification, sterilization, and concentration of suspensions.

PERPENDICULAR VS. CROSS FLOW FILTRATION



Advantages of Cross-Flow Filtration

- By continuously sweeping the membrane surface, **CFMF prevents the formation of a thick, high-resistance filter cake**. This maintains a higher permeate flux (J) for a longer period.
- The **retentate remains a fluid, pumpable slurry**. This is crucial for downstream processing. In conventional filtration, **the cake is a solid "mud" that must be scraped or washed off**.
- The **concentration of the retentate can be easily controlled** by adjusting the permeate removal rate (or the recycle ratio). You can produce a dilute or a highly concentrated slurry as needed.

Process Flow Diagram - Basic Cross-Flow System

Simple Pump Recirculation Loop: This is the system typically used for batch processing or development rigs.

Operational Principle:

- Process feed is pumped from the tank and across the membrane in the module at an appropriate velocity.
- Partially concentrated retentate is recycled into the tank for further processing.
- Permeate is stored or discarded as required.

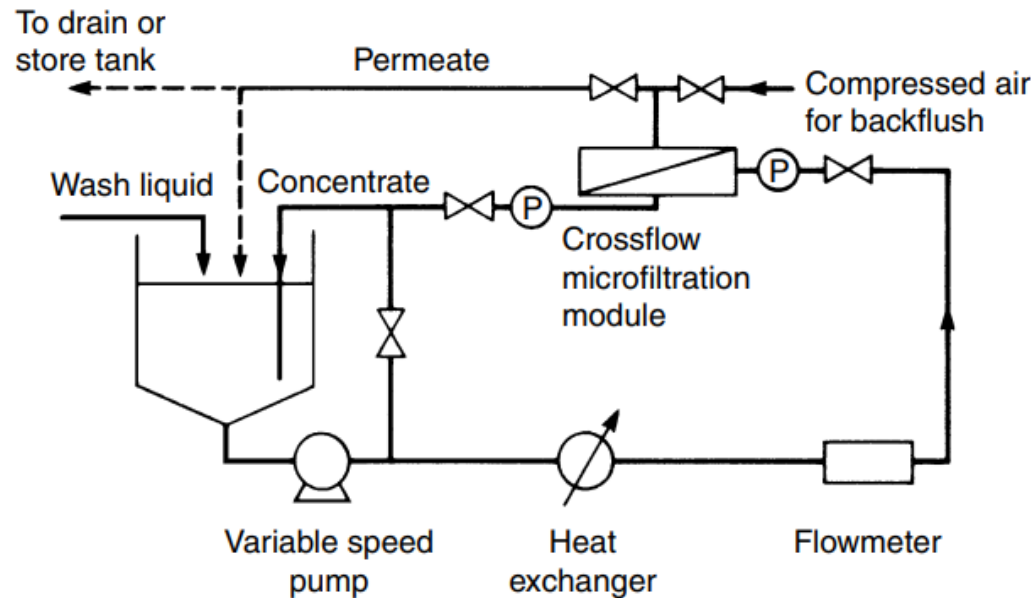


Figure 8.5. Flow diagram for a simple cross-flow system⁽⁴⁾

Process Flow Diagram - Basic Cross-Flow System

Diafiltration - Product Washing

- **Need for Washing:**

- In cross-flow filtration applications, product washing is frequently necessary.

Diafiltration Process:

- Wash water is added to the tank at a rate equal to the permeation rate.

Objective:

- To wash out unwanted small molecules or contaminants from the retained product.
- Maintains constant volume while purifying the retentate.

Membrane Fouling .

What is Fouling?

- Blocking of the membrane surface and pores by the particulate materials.

Types of Fouling: cake fouling, pore blocking, adsorption, gel formation, biofouling

Consequences:

- Reduced flux.
- Increased resistance.
- Higher operating costs.

Ultrafiltration

- Ultrafiltration is one of the most widely used of the pressure-driven membrane separation processes.
- Particle size $<0.1\ \mu\text{m}$ – 5 nm
- **Solute Retention:**
 - Solutes retained or rejected by ultrafiltration membranes are those with molecular weights of 10^3 or greater.
 - This depends mostly on the MWCO (Molecular Weight Cut-Off) of the membrane chosen.
- **Permeable Components:** The process liquid, dissolved salts, and low molecular weight organic molecules (500–1000 kg/kmol) generally pass through the membrane.

Operating Conditions

- **Transmembrane Pressure:** The pressure difference applied across the membrane is usually in the range 0.1–0.7 MN/m² (0.1–0.7 MPa).
- **Permeation Rates:** Membrane permeation rates are typically 0.01–0.2 $\text{m}^3/\text{m}^2\ \text{h}$.
- **Operating Mode:** In industry, ultrafiltration is always operated in the cross-flow mode.

Ultrafiltration

Concentration Polarisation

- **Phenomenon:** The separation of process liquid and solute that takes place at the membrane during ultrafiltration gives rise to an increase in solute concentration close to the membrane surface.
- **Location:** This is termed **concentration polarisation** and takes place within the boundary film generated by the applied cross-flow.

Back-Diffusion and Steady State

- **Driving Force for Back-Diffusion:** With a greater concentration at the membrane, there will be a tendency for solute to diffuse back into the bulk feed according to **Fick's Law** (the diffusive flux is proportional to the concentration gradient).
- **Steady State Condition:** At steady state, the rate of back-diffusion will be equal to the rate of removal of solute at the membrane, minus the rate of solute leakage through the membrane.
- **Mathematical Expression:**

$$J(C - C_p) = -D \frac{dC}{dy}$$

Where:

J = Membrane flux

C = Solute concentration at membrane surface

C_p = Solute concentration in permeate

D = Diffusivity of solute

dC/dy = Concentration gradient in the boundary layer

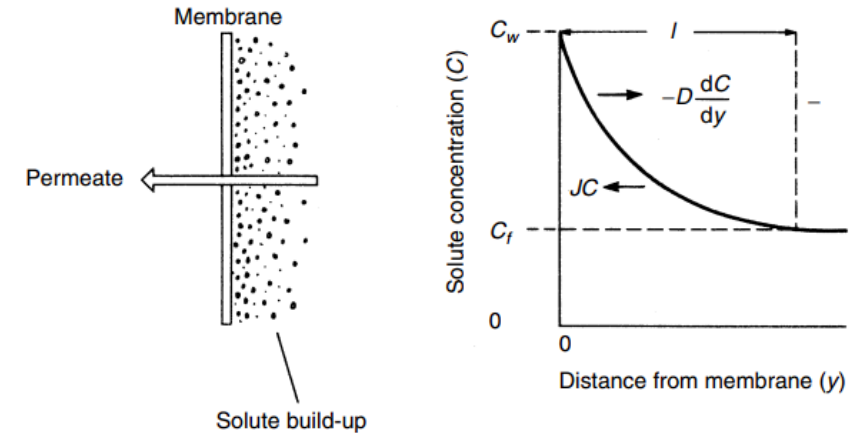


Figure 8.7. Concentration polarisation at a membrane surface

Ultrafiltration

Integration of the Mass Balance Equation

Starting Equation:

$$J(C - C_p) = -D \frac{dC}{dy}$$

Rearrange:

$$- \int_{C_w}^{C_f} \frac{dC}{C - C_p} = \frac{J}{D} \int_0^l dy$$

C_f = Solute concentration in feed

C_w = Solute concentration at wall

Integrate:

$$\ln \left(\frac{C_w - C_p}{C_f - C_p} \right) = \frac{Jl}{D}$$

Final Form:

$$\frac{C_w - C_p}{C_f - C_p} = \exp \left(\frac{Jl}{D} \right)$$

Interpretation: The exponential relationship shows that even a modest flux (J) can lead to a significant increase in wall concentration.

Ultrafiltration

Complete Solute Rejection

Assumption: The membrane completely rejects the solute.

$$R = 1 \text{ (Rejection coefficient = 1)}$$

$$C_p = 0 \text{ (No solute passes through)}$$

Equation Simplifies to:

$$\frac{C_w}{C_f} = \exp\left(\frac{Jl}{D}\right)$$

Polarisation Modulus:

The ratio C_w/C_f is known as the **polarisation modulus**.

It quantifies the extent of concentration build-up at the membrane surface.

Physical Meaning:

If $C_w/C_f = 3$, the concentration at the wall is three times that in the bulk feed.

Higher modulus = More severe polarisation = Greater fouling potential.

Case Study: Dairy Industry - Concentration of Whey Protein

Process: A cheese manufacturing plant produces large volumes of whey as a by-product. Whey contains valuable whey proteins with molecular weights ranging from ~14,000 to 150,000 Da, along with lactose (MW ~342 Da), minerals, and water. The plant wants to recover and concentrate the whey proteins for use in sports nutrition and food products.

Application of Ultrafiltration:

- The whey is processed through a cross-flow ultrafiltration system using a membrane with a MWCO of 10,000 Da.
- **Retained:** Whey proteins (MW $\geq$ 10,000 Da).
- **Passes Through:** Water, lactose, and minerals.

The Role of Concentration Polarisation:

- As the whey is concentrated, proteins accumulate at the membrane surface. This increases the local concentration, leading to a gel layer formation.
- The gel layer adds significant resistance, reducing the flux. The plant observes that the flux declines over time, and simply increasing the pressure does not restore it—it actually makes it worse by compressing the gel layer.

Solution:

- The plant **optimizes the cross-flow velocity** to increase shear and reduce the boundary layer thickness, promoting back-diffusion of proteins into the bulk feed.
- Periodic **backflushing and cleaning-in-place** (CIP) are employed to remove the gel layer.
- The **diafiltration process** (adding water to the retentate) is used to wash out additional lactose and minerals, purifying the protein concentrate.

Result: The plant recovers 95%+ of the whey proteins, producing a high-value whey protein concentrate. The concentration polarisation is managed through high cross-flow velocities and regular cleaning, making the process economically viable.

Reverse Osmosis

Osmosis:

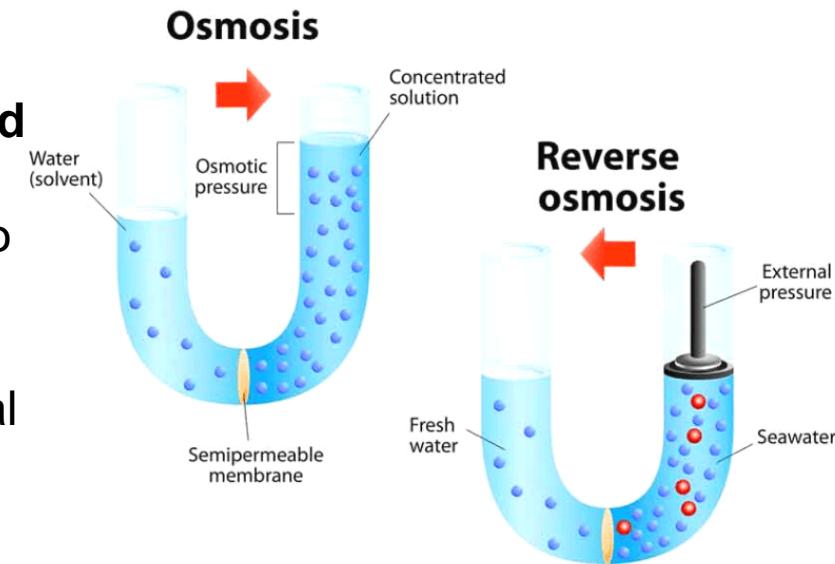
What is Osmosis?

- The spontaneous diffusion of a solvent (typically water) across a semi-permeable membrane.
- The membrane is permeable to the solvent but **impermeable** to the solute.
- **Direction:** From the **less concentrated** side to the **more concentrated** side.
- Water molecules are "more active" on the dilute side and naturally try to equalize the concentration on both sides by diluting the concentrated solution.

Key Concept: The transport of solvent will continue until the chemical potential (how strongly a substance "wants" to move, react, or change its state) of the solvent is equal on both sides.

Example: Tea bag process

- The water moves *into* the tea bag, dissolving the flavors and colors.
- The tea leaves (solutes) stay inside the bag.
- The tea bag is like a **semipermeable membrane**



Reverse Osmosis

Osmotic Pressure (π)

- The pressure that must be applied to the **concentrated** solution to stop the flow of solvent (i.e., to achieve osmotic equilibrium).
- In other words, it's the pressure required to equalize the solvent activities on both sides of the membrane.

Example:

- Imagine pure water on one side of a membrane and a salt solution on the other.
- Initially, water flows into the salt solution. This flow causes the liquid level on the salt side to rise.
- The height difference (h) creates a hydrostatic pressure. The pressure at equilibrium is the **osmotic pressure** (π).

Mathematical Representation:

For dilute solutions, the van 't Hoff equation applies:

$$\pi = iMRT$$

Where:

i = van 't Hoff factor (number of ions per molecule, e.g., $\text{NaCl} \rightarrow i \approx 2$)

M = Molar concentration of the solute

R = Ideal gas constant ($8.31 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$)

T = Absolute temperature

Reverse Osmosis

"Reverse" Concept:

- If we apply a pressure **greater than** the osmotic pressure on the concentrated side, the flow of solvent reverses.
- The solvent is forced from the **concentrated** solution to the **dilute** side (or pure solvent side).
- This is **Reverse Osmosis**.

The Process Flow:

- **High-Pressure Feed:** Concentrated solution (e.g., seawater) is pumped at high pressure (typically 50-100 atm).
- **Permeate:** Pure solvent (water) passes through the membrane.
- **Retentate/Concentrate:** The remaining, highly concentrated solution (brine) is discharged.

Normal Osmosis vs. Reverse Osmosis:

- **Osmosis:** Water moves from dilute → concentrated. No external pressure needed (except to stop it).
- **Reverse Osmosis:** Water moves from concentrated → dilute. **Requires** external pressure $> \pi$.

Reverse Osmosis

Energy Considerations and Advantages

Why RO is Energy-Efficient:

- **No Phase Change:** Unlike distillation or evaporation, RO does not require vaporization. No latent heat of vaporization needs to be supplied or removed. This saves enormous amounts of energy.
- **Room Temperature Operation:** RO can be conducted at ambient temperature. This is critical for heat-sensitive materials (e.g., food, pharmaceuticals, biologics).

Energy Source:

- The energy for separation comes from the **work of pressurizing the feed**.
- This is mechanical work performed by a high-pressure pump.

Membrane Materials - The Historical Challenge

Early Membranes (Dense Films):

- Scientists (e.g., Reid and Breton) experimented with dense, homogeneous films of cellulose acetate.
- **Performance:**
 - High salt rejection (95-98%).
 - However, water flux was **impractically low**.
- **Trade-off:** Higher acetyl content in cellulose acetate gave even higher rejection but even lower flux.

Problem with Dense Membranes:

- To achieve high salt rejection, the membrane must be dense (non-porous). But a thick dense layer offers massive resistance to water flow, giving a very low flux (J).
- It's like trying to filter water through a solid sheet of plastic. It works but is extremely slow.

Reverse Osmosis

Breakthrough - Asymmetric Membranes

Loeb-Sourirajan Discovery (1960s): This revolutionized RO and made it a practical, industrial-scale process.

Structure:

- An **asymmetric membrane** has a two-layer structure made from the same polymer:
 - **A very thin, dense "skin" layer** (less than 1 μm thick). This provides the high salt rejection.
 - **A thick, highly porous substructure** (100-200 μm). This provides mechanical support with very low resistance to water flow.
- The ultra-thin selective layer is all that matters for rejection.
- The porous substructure acts only as a support.
- **Result:** High salt rejection from the skin layer, but high flux because the resistance path through the dense layer is very short.

Membrane Geometries and Materials

Geometries (Membrane Configurations):

- **Flat Sheets:** Stacked in plate-and-frame or spiral-wound modules (most common for RO).
- **Hollow Fibers:** Thousands of tiny, hair-thin fibers bundled together. Offer very high surface area per unit volume.
- **Tubular Membranes:** Membranes coated on the inside of porous pipes

Materials:

- **Cellulose Acetate (CA):** The original material. Cost-effective, but sensitive to pH, temperature, and chlorine.
- **Aromatic Polyamides (e.g., DuPont's "Permasep"):** polyamide thin-film composites

What is a Module?

- The basic, self-contained unit of membrane equipment.
- Contains the membrane and the necessary supports, spacers, and collection systems.
- Essentially a "plug-and-play" component for building a membrane plant.

Why Modules?

- **Standardization:** Modules are manufactured to standard sizes and performance specifications.
- **Scalability:** You can achieve the required membrane area by connecting modules together.
- **Maintainability:** Damaged modules can be isolated and replaced without shutting down the entire plant.
- **Typical Module Area:** 1 – 20 m² of membrane surface area per module.

How They Are Connected:

- **Parallel:** Increases flow capacity (more permeate).
- **Series:** Increases concentration factor (more retentate concentration).

Module Type 1 - Tubular Modules

Construction:

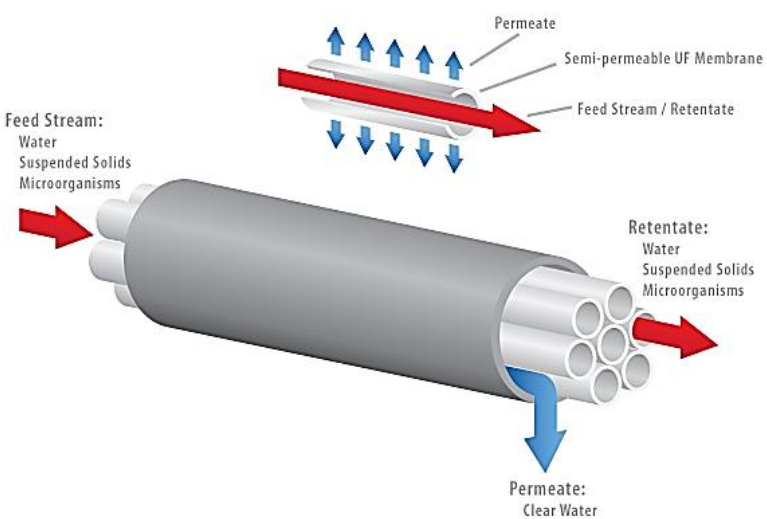
- The membrane is cast on the **inside** of a porous support tube (often made of a ceramic or polymer).
- This tube is typically housed in a perforated stainless steel pipe for mechanical protection.
- A module contains a cluster of these tubes (typically 10–25 mm diameter, 1–6 m long) in a stainless steel shroud.

Operation:

- Feed is pumped through the tubes at high velocity.
- Reynolds numbers are typically **> 10,000** (turbulent flow).
- Permeate passes radially outward through the membrane and support, collecting in the shroud.

Flow Regime: Turbulent (designed for high shear to control fouling).

Advantages	Disadvantages
Excellent for high-solids feeds	Low membrane area per unit volume
Easy to clean (can use sponge balls)	High volumetric hold-up (large inventory)
Well-established technology	Higher energy consumption (due to high flow rates)
Can handle abrasive slurries	Higher capital cost per m ² of membrane



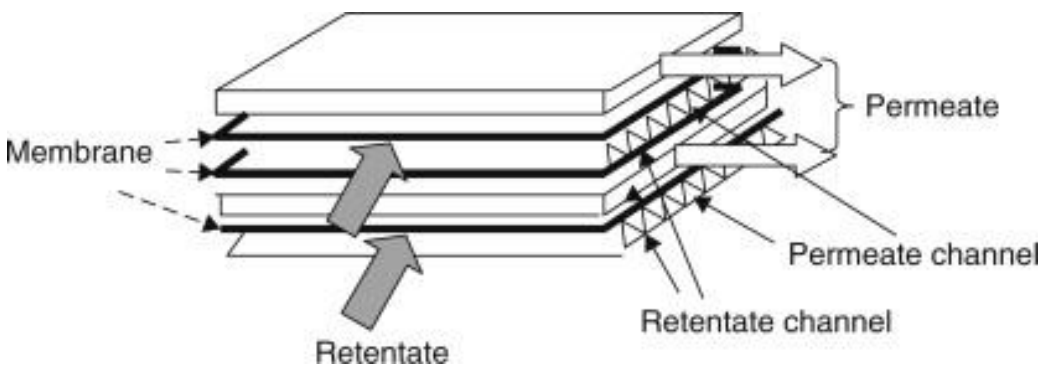
Construction:

- Similar in concept to a **filter press**.
- Consists of a series of annular membrane discs placed on both sides of a support plate (e.g., polysulphone).
- Support plates provide channels for permeate collection.
- Spacer plates with central and peripheral holes separate the membrane/support "sandwiches."

Operation:

- Feed flows over the membrane surfaces through the spacer plates.
- The flow is typically **laminar**.
- Permeate is collected from each membrane pair.

Advantages	Disadvantages
High membrane area density	Entire stack dismantling required for membrane replacement
Easy to identify damaged membranes	Difficult to clean thoroughly
Relatively low hold-up volume	Laminar flow leads to higher concentration polarization
Good for delicate or shear-sensitive materials	



Module Type 3 - Spiral-Wound Modules

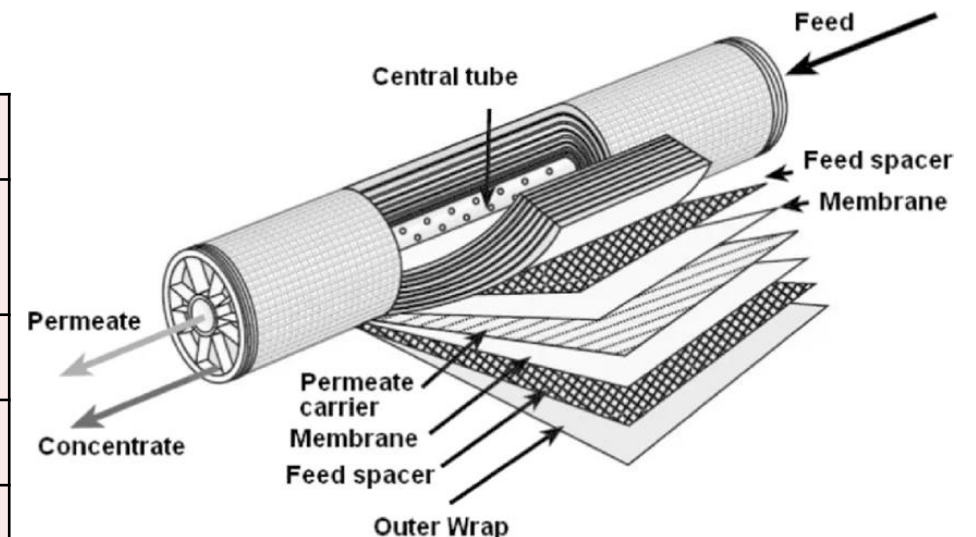
Construction:

- Multiple flat membranes separated by **turbulence-promoting mesh spacers**.
- The layers are sealed together and rolled around a central perforated collection tube.
- The resulting cylindrical module is installed inside a pressure tube.
- Standard size: ~0.1 m diameter, ~0.9 m length, containing ~**5 m²** of membrane area.
- Up to **6 modules** can be connected in series in a single pressure tube.

Operation:

- Feed enters the pressure tube and flows axially through the narrow feed channels.
- Permeate spirals radially inward through the membrane layers to the central collection tube.

Advantages	Disadvantages
Best space utilization (high area/volume ratio)	Prone to fouling (narrow channels)
Lower capital cost per m ² of membrane	Difficult to clean (cannot use mechanical cleaning)
Good for large-scale applications	Limited to relatively clean feeds
Widely used in RO and NF applications	Feed spacer can cause pressure drop and fouling

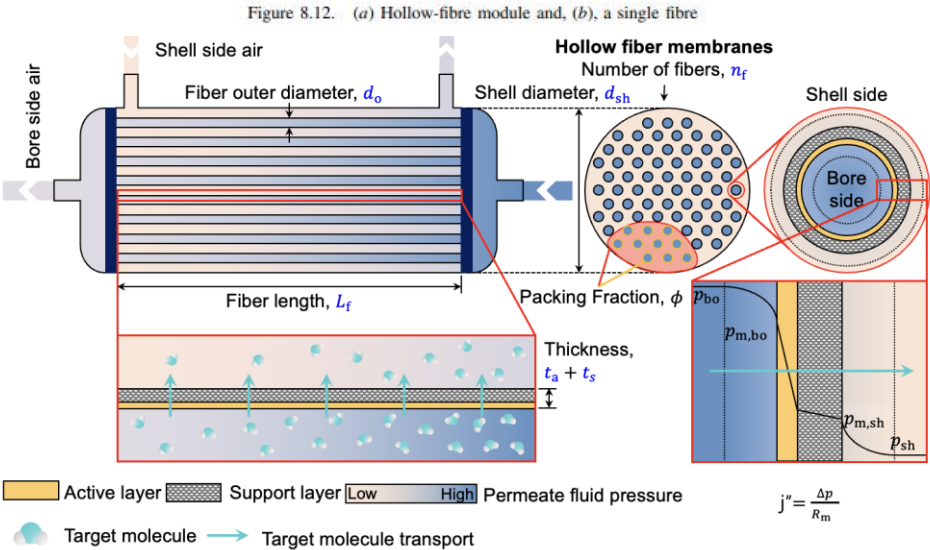
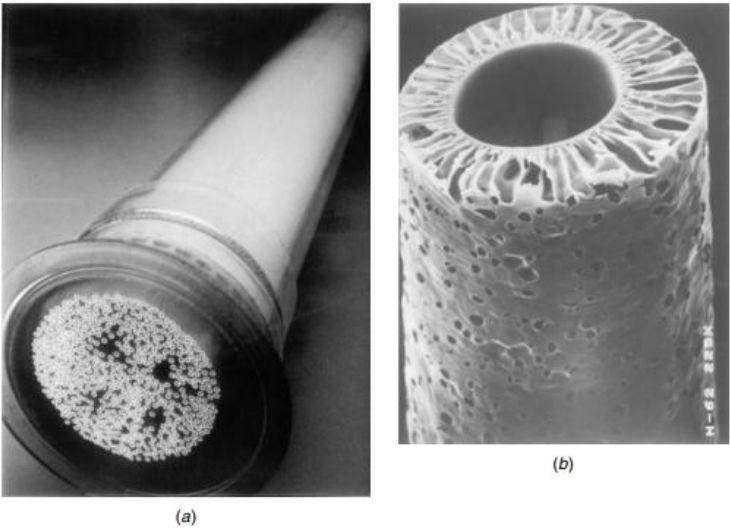


Construction:

- Consists of **bundles of fine fibres** (0.1–2.0 mm diameter) sealed in a tube (shell-and-tube configuration).
- A single UF module can contain up to **3000 fibres** and be ~1 m long.
- RO modules contain more, finer fibres for higher area density.

Mode	Feed Flow	Permeate Flow	Pressure	Application
Outside-In (Conventional RO)	Outside the fibres	Radially inward	Very High	Reverse Osmosis (desalination)
Inside-Out (Biotech/UF)	Down the centre of the fibres	Radially outward	Low (< 0.2 MPa)	MF/UF for cell cultures

Advantages	Disadvantages
Highest membrane area density (most compact)	Very prone to fouling (narrow channels)
Self-supporting (no support structure needed)	Very difficult to clean
Low hold-up volume	Feed flow channels can be < 0.1 mm wide
Can withstand very high pressures (outside-in mode)	



The Need for Different Configurations:

- The simple batch recirculation system is suitable for small-scale operations.
- For large-scale industrial production, we need continuous, high-throughput systems.
- The choice of configuration affects: **Product concentration, Membrane area required, Residence time, and Operating costs.**

Three Main Configurations:

- **Batch Recirculation** (Small scale, development rigs)
- **Feed and Bleed** (Continuous, industrial scale)
 - Single-stage
 - Multiple-stage (Multi-stage)
- **Continuous Single Pass** (Very large scale, RO desalination)

Objective: To achieve continuous production of a concentrated retentate.

Operating Principle:

- **Start-up:** The retentate is totally recycled (like a batch system) until the desired solute concentration is reached in the loop.
- **Steady-State:** Once the target concentration is achieved, a fraction of the loop is continuously "bled off" as product (retentate).
- **Feed Control:** Fresh feed is pumped into the loop at a rate equal to the sum of (Permeate flow + Bleed/Retentate flow).

Single-Stage Feed and Bleed

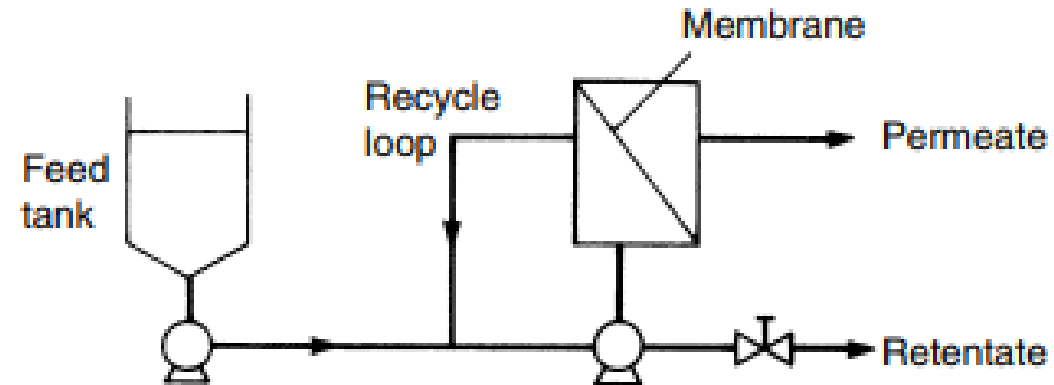
- **Advantage:** Final concentration product is continuously available.
- **Disadvantage:** The entire loop operates at the **final, highest concentration**. This means flux (J) is at its lowest throughout the system, requiring a **larger membrane area** for the same throughput.

Multiple-Stage Feed and Bleed

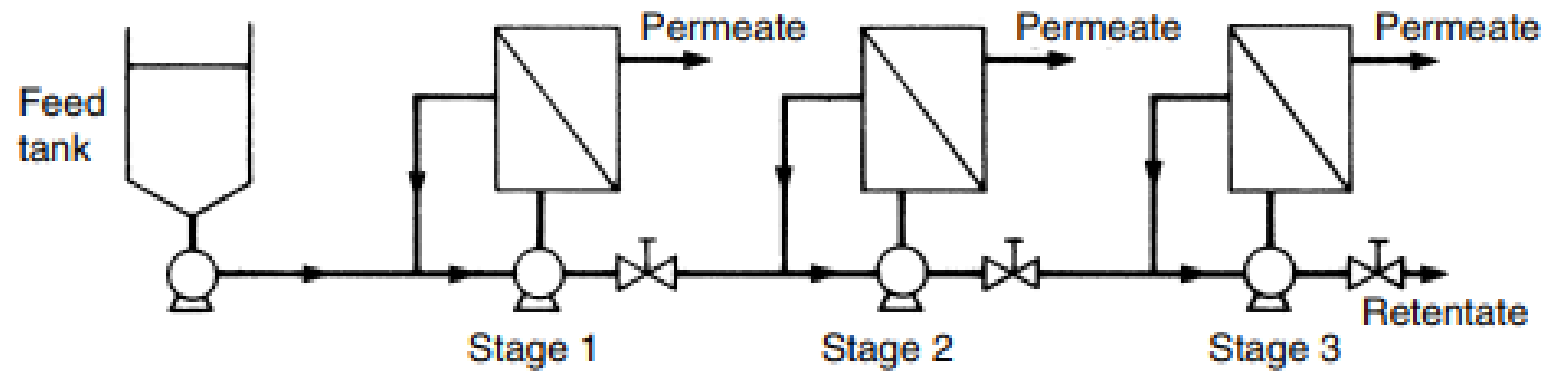
Solution: Stages are operated in series.

- **Benefit:** Only the final stage operates at the highest concentration (lowest flux). Earlier stages operate at lower concentrations (higher flux).
- **Result:** Total membrane area required is **significantly less** than a single-stage system.
- **Typical Design:** A minimum of three stages is usually required.
- **Additional Advantages:**
 - Lower residence time and hold-up volume.
 - Less frequent sterilization needed (important in biotech).

Configuration 1: Feed and Bleed System



Single stage



Multistage

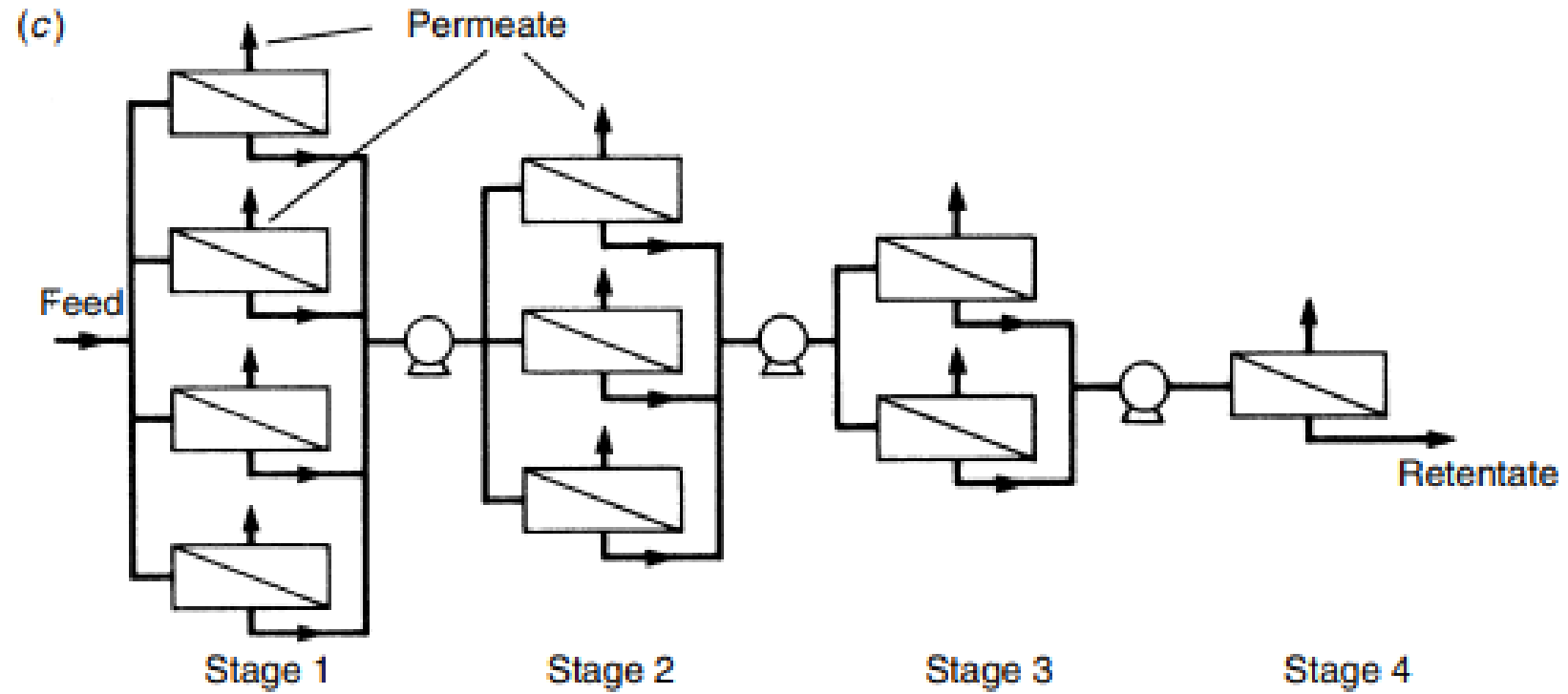
Continuous Single Pass Concept:

- Feed flows through several membrane stages in series, with **no recycle**.
 - **Concentration Profile:** The feed concentration increases gradually along the length of the stages. It only reaches its final concentration at the very last stage.
 - **Advantage:** Very low residence time. Minimal product hold-up.

Limitations:

- Requires very high cross-flow velocities to control concentration polarization (to keep particles swept away).
- Must be applied on a **very large scale** (e.g., $>0.1 \text{ m}^3/\text{min}$ feed per final stage) OR have a low overall concentration factor.
- **Primary Application:** Large-scale Reverse Osmosis (RO) desalination plants.
- **Not Suitable For:** Biotechnological applications (due to scale requirements).

Configuration 2: Continuous Single Pass



What is Membrane Fouling?

- The accumulation of materials (particles, colloids, macromolecules, salts) on the membrane surface or within its pores.
- This is the single biggest limitation to the wider adoption of membrane processes.

Consequences of Fouling:

- **Continuous Decline in Permeation Rate:** The flux (J) drops continuously during operation.
- **Increased Rejection of Low Molecular Weight Solutes:** As pores become blocked, the effective pore size decreases, causing the membrane to retain smaller molecules than intended.
- **Blocking of Flow Channels:** In severe cases, the entire flow channels within the membrane module can become completely blocked, halting the process.

The Two Mechanisms of Fouling

1. Pore Blocking / Pore Narrowing:

- Particles or molecules become lodged *inside* the membrane pores, physically blocking them or reducing their effective diameter.
- This is often irreversible and reduces the membrane's permeability.

2. Surface Layer Formation (Cake/Gel Layer):

- Materials deposit and accumulate on the membrane surface, forming a layer that grows thicker over time.
- This adds to the resistance (R_c) as described in the General Membrane Equation.

Rate and Extent of Fouling

- **Rapid Initial Drop:** A steep decline from $J_{pure\ water}$ to 30-10% within minutes.
- **Gradual Decline:** A slower, continuous decline over hours/days.
- **Labeled regions:** "Initial Rapid Fouling" and "Gradual Fouling / Cake Growth."

Fouling Curve

Start-Up: A new or freshly cleaned membrane has its maximum permeation rate (Pure Water Flux).

Rapid Initial Decline (First Few Minutes):

- For **Ultrafiltration (UF)**: Flux drops to **30–10%** of the pure water flux within minutes.
- For **Microfiltration (MF)**: The decline can be **even more extreme**.
- **Why so fast?** Initially, the largest pores are quickly blocked by the largest particles or aggregates in the feed. This happens almost immediately.

Gradual Decline (Long-Term Operation):

- After the initial sharp drop, the flux continues to decrease, but at a much slower, more gradual rate.
- **Why?** This is due to the slow accumulation of a cake/gel layer on the membrane surface and the progressive blocking of smaller pores.

Implications for Design:

- You cannot design a membrane system based on pure water flux. The operating flux will be a fraction of that value.
- The initial rapid fouling determines the "steady-state" flux, while the gradual decline dictates the cleaning frequency.

1. Membrane Selection

- The extent of fouling depends heavily on the **nature of the membrane** (material, pore size, surface charge, hydrophobicity).
- **Hydrophilic membranes** (water-loving) tend to foul less than hydrophobic membranes because they have less affinity for organic foulants.
- **Careful Choice:** Matching the membrane chemistry to the feed is the first and most important line of defense.

2. Module Design & Hydrodynamics: The module must provide suitable hydrodynamic conditions for the specific application.

- **Cross-flow velocity:** High velocities create shear that sweeps away deposited particles
- **Pulsated Feed Flows:** Intermittent or pulsating flow disrupts the boundary layer.
 - **Non-Planar Membrane Surfaces:** Complex geometries create turbulence.
 - **Applied Perturbations:** Continuous or pulsated **electric fields** can repel charged foulants.

3. Feed Pretreatment: Modifying the feed before it reaches the membrane can dramatically reduce fouling.

Application	Pretreatment Method	Purpose
Desalination (RO)	Coagulation, Filtration, Anti-scalants	Remove suspended solids, prevent salt precipitation
Biotech / Food	Prefiltration	Remove large particles before the main membrane
	Pasteurization	Destroy bacteria that could colonize the membrane
	pH / Ionic Strength Adjustment	Prevent protein precipitation at the membrane surface

Cleaning Strategies: When Fouling Occurs

1. Backflushing / Backwashing:

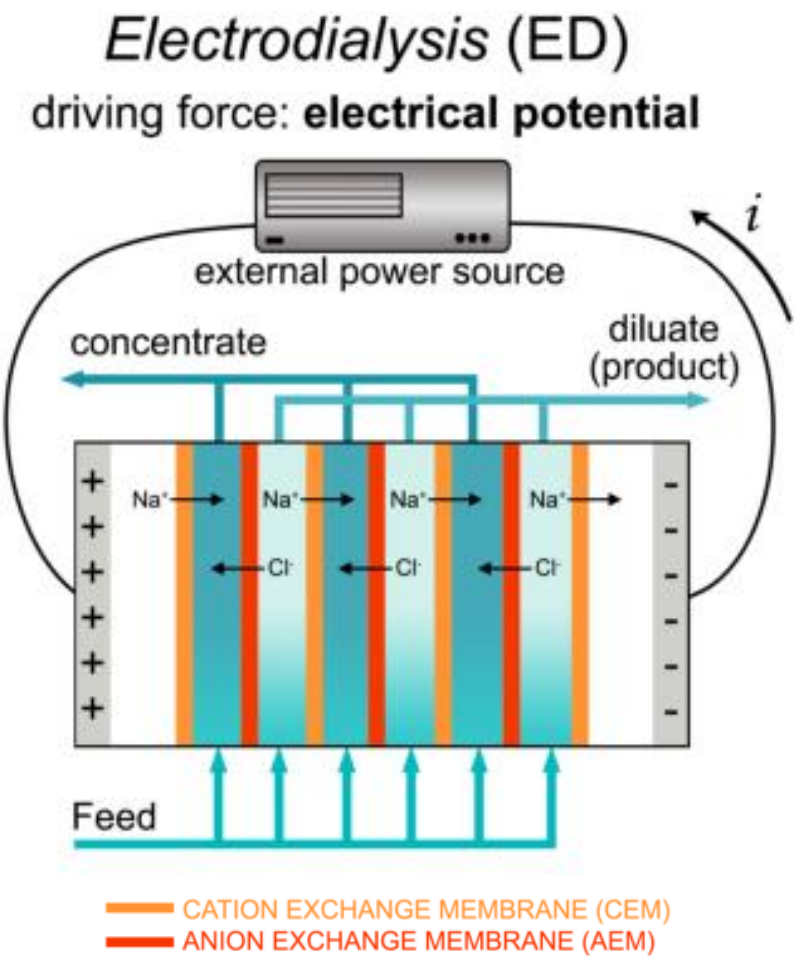
- Reversing the flow of permeate through the membrane to dislodge foulants.
- **Effectiveness:** Can substantially restore the permeation rate.
- **Limitation:** Seldom totally effective. It removes loose cake layers but not strongly adsorbed materials or pore blockages.

2. Chemical Cleaning (The Inevitable Step):

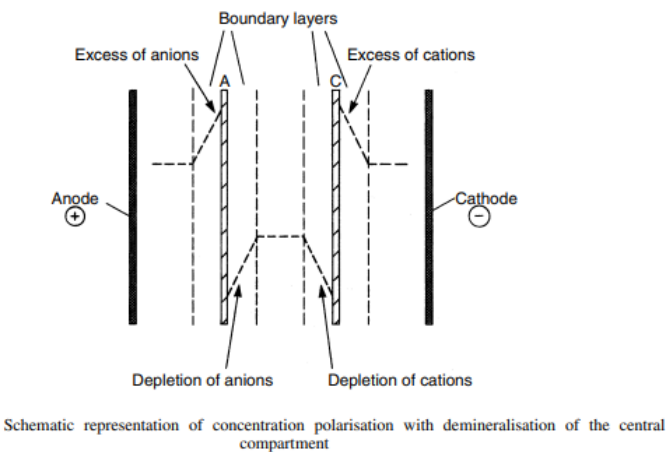
- Eventually, backflushing is insufficient, and chemical cleaning is required.
- **Drawback:** This requires interrupting the separation process, leading to downtime.

Electrodialysis (ED)

- **Definition:** An electrically driven membrane separation process.
- **Unlike pressure-driven processes** (RO, UF, MF), ED uses an **electrical potential** as the driving force.
- **Mechanism:** Ions migrate through ion-selective membranes under the influence of an applied electric field.



Membrane Type	Function	Permeability
Cation Exchange Membrane (CEM)	Allows cations (positive ions) to pass through	Transports Na^+ , Ca^{2+} , K^+
Anion Exchange Membrane (AEM)	Allows anions (negative ions) to pass through	Transports Cl^- , SO_4^{2-} , NO_3^-



Electrodialysis Stack Configuration and Operation

Stack Configuration Membrane Array:

- Cation exchange membranes (C) and anion exchange membranes (A) are **alternated** in a parallel array.
- This creates **solution compartments** between each pair of membranes.
- Cations moving toward cathode, anions toward anode.
- **Compartment Thickness:**
 - Each solution compartment is approximately **1 mm thick**.
 - This thin gap minimizes electrical resistance and enhances ion transport.
- **Stack Size:**
 - A single membrane stack typically contains **100–400 membranes**.
 - Each membrane has an area of **0.5–2.0 m²**.
 - Total membrane area in a single stack can be 50–800 m².

How the Process Works?

Feed Introduction: The process feed (saltwater) is pumped through the solution compartments.

Applying Electrical Potential:

- An electrical potential is applied to the electrodes.
- **Anode (+):** Attracts anions (negative ions).
- **Cathode (-):** Attracts cations (positive ions).

Ion Transport Across Membranes:

- **Cations (e.g., Na^+):** Transported toward the cathode.
 - They can cross **cation exchange membranes**.
 - They **cannot** cross anion exchange membranes.
- **Anions (e.g., Cl^-):** Transported toward the anode.
 - They can cross **anion exchange membranes**.
 - They **cannot** cross cation exchange membranes.

Result:

- **Depleted Compartments:** Ions are removed → demineralized (product) solution.
- **Concentrated Compartments:** Ions accumulate → concentrated brine (byproduct).

Spacer Designs and Hydrodynamics in ED

Role of Spacers

- **Purpose:** Keep the membranes apart, defining the **flow channels** for the process feed.
- **Thickness:** Maintain the ~1 mm compartment gap.
- **Additional Function:** Act as **turbulence promoters** to reduce concentration polarization.

Feature	Tortuous Path Spacer	Sheet Flow Spacer
Flow Path	Long narrow channels with multiple 180° bends	Straight path from entrance to exit
Channel Length-to-Width Ratio	~100:1	Straight, no bends
Turbulence Promotion	High (due to bends and screens)	Moderate (due to spacer screens)
Pressure Drop	Higher	Lower
Residence Time	Longer	Shorter
Application	When high mass transfer is needed	When low pressure drop is critical

Turbulence Promotion and Concentration Polarization

Why Turbulence Matters:

- In ED, concentration polarization occurs at the membrane interfaces
- **Turbulence promoters** (the spacer screens) mix the solution near the membrane surface.
- This mixing **reduces the boundary layer thickness**, enhancing ion transport to/from the membrane.

Difference from Pressure-Driven Processes:

- In RO/UF, concentration polarization affects **solvent flux** (osmotic pressure effects).
- In ED, concentration polarization affects **ion transport** and **limiting current density**.
- The effect is "rather different" but equally important.

Design Considerations

Tortuous Path Spacers:

- Provide better mass transfer (higher turbulence).
- But higher pressure drop means more pumping energy.
- Used when high demineralization rates are needed.

Sheet Flow Spacers:

- Provide lower pressure drop.
- But lower mass transfer means thicker boundary layers.
- Used when feed flow rates are high or pressure drop must be minimized.

Major Industrial Applications of ED

Desalination of Seawater and Brackish Water:

- Primary application for producing drinking water.
- Particularly effective for brackish water (lower salt concentration than seawater).
- More energy-efficient than RO for low to moderate salinity feeds.

Sodium Chloride (Table Salt) Production:

- **The principal method in Japan.**
- Production exceeds **1 million tonnes per annum** (10^6 tonne/year).
- ED concentrates seawater to produce high-purity NaCl brine, which is then crystallized.

Other Applications

- Food industry: Demineralization of whey, sugar, and fruit juices.
- Chemical industry: Recovery of acids and bases from waste streams.
- Pharmaceutical: Production of ultra-pure water.

A municipality in the Middle East relies on brackish groundwater for its water supply. The groundwater has a total dissolved solids (TDS) concentration of 5,000 ppm (about 0.085 kmol/m^3). The municipal water standard requires TDS below 500 ppm. The existing reverse osmosis plant is struggling with high energy costs because the osmotic pressure of brackish water, while not as high as seawater, is still significant.

- **The ED Solution:**

The municipality decides to install an electrodialysis plant as an alternative to RO. A pilot study was conducted using a small ED stack with the following results:

- Feed concentration: 0.085 kmol/m^3
- Measured limiting current density: 340 A/m^2
- Polarization parameter: $340/0.085 = 4,000 \text{ A}\cdot\text{m}^2/\text{kmol}$

Plant Design: The full-scale plant needs to treat $10,000 \text{ m}^3/\text{day}$ of brackish water, reducing TDS from 5,000 to 500 ppm (90% rejection).

Calculations:

- Using the polarization parameter, the limiting current density at the feed concentration is $4,000 \times 0.085 = 340 \text{ A/m}^2$.
- Operating at 60% of limiting: $0.60 \times 340 = 204 \text{ A/m}^2$.
- Total current required: $10,000 \text{ m}^3/\text{day} = 417 \text{ m}^3/\text{hour}$. With a stack current of 100 A per stack, they need 417 stacks.
- Total power: $417 \text{ stacks} \times 150 \text{ V} \times 100 \text{ A} = 6.25 \text{ MW}$.

Results:

- The ED plant produces $10,000 \text{ m}^3/\text{day}$ of drinking water.
- Energy consumption: $6.25 \text{ MW} \times 24 \text{ h} / 10,000 \text{ m}^3 = 15 \text{ kWh/m}^3$.
- Compare to RO: RO would require $\sim 5 \text{ kWh/m}^3$ for brackish water, but with higher chemical costs for anti-scalants and more frequent membrane replacement.
- ED has lower chemical costs and longer membrane life (ion exchange membranes are more robust than RO membranes).
- Overall operating cost: ED is 20% cheaper than RO for this specific brackish water application.

Background: Japan has limited land for solar salt production and historically relied on imported salt. In the 1950s, Japan began developing electrodialysis technology to produce salt from seawater, which is abundant around the Japanese islands.

- **The Process:** The ED plant takes seawater (3.5% NaCl) and concentrates it to ~20% NaCl using electrodialysis. The concentrated brine is then evaporated to crystallize pure sodium chloride.
- **Scale of Operation:** Japan produces **over 1 million tonnes of salt per year** using this method. One of the largest producers operates a plant with 1,000 ED stacks, each containing 400 membranes of 2 m² area.

Calculations:

- Total membrane area: 1,000 stacks × 400 membranes × 2 m² = 800,000 m² of membrane area.
- Each stack operates at 100 A, 150 V → 15 kW per stack.
- Total power: 1,000 stacks × 15 kW = 15 MW.
- Annual energy consumption: 15 MW × 24 h × 365 days = 131,400 MWh/year.
- Salt production: 1,000,000 tonnes/year.
- Energy per tonne of salt: 131,400 MWh / 1,000,000 tonnes = 131.4 kWh/tonne.

Challenges:

- **Membrane fouling:** Seawater contains calcium, magnesium, and sulfate ions that can precipitate on the membranes during concentration.
- **Concentration polarization:** At high concentrations, the limiting current density is reached more quickly, limiting the maximum concentration that can be achieved.

Solutions:

- **Feed pretreatment:** Seawater is filtered and chemically treated to remove calcium and magnesium (e.g., by adding sodium carbonate to precipitate CaCO₃).
- **Staged operation:** The ED process is operated in multiple stages, with each stage operating at a different current density to avoid exceeding the limiting value.

Result: Japan is now self-sufficient in salt production, and the ED process has become the standard for salt production in the country. The salt produced is of high purity (99.5%+ NaCl), suitable for table salt and industrial uses.

Problem Pervaporation Solves

Two Difficult Separation Categories:

- **Azeotropic Mixtures:** Where liquid and vapour compositions are identical at a certain composition (e.g., ethanol-water at 95.6% ethanol).
- **Close-Boiling Mixtures:** Where components have very similar boiling points (e.g., benzene and cyclohexane, boiling points differ by only 0.6°C).

Traditional Solutions are Costly:

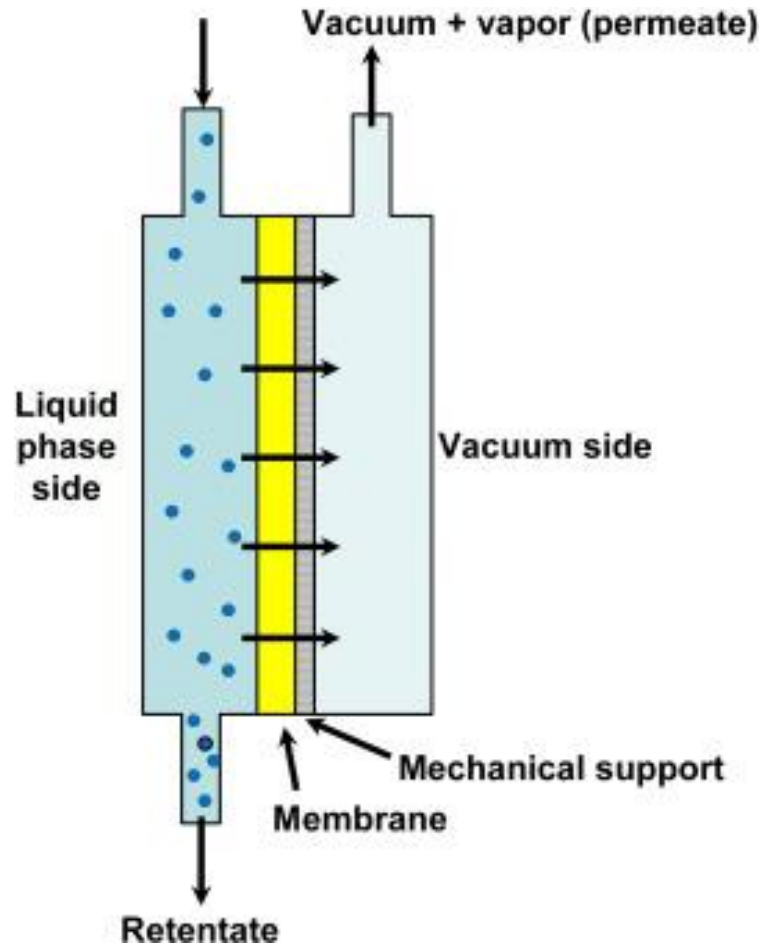
- **Azeotropic Distillation:** Requires adding a third component (entrainer).
- **Extractive Distillation:** Uses a solvent to alter relative volatility.
- **Pressure Swing Distillation:** Requires high pressure and energy.

Pervaporation Offers an Alternative: A membrane-based process with potentially lower energy consumption.

What Makes Pervaporation Unique?

- **Phase Change:** The permeate undergoes a **liquid-to-vapour phase change** during transport.
- **Driving Force:** Achieved by **lowering the activity** of permeating components on the permeate side.
 - This is done by maintaining **partial pressures** below the **saturation vapour pressures** of the permeating components.
- **How?** A **vacuum** is applied on the permeate side

Pervaporation



1. **Feed Tank:** Contains the liquid mixture to be separated.
2. **Feed Pump:** Pressurizes the feed to flow through the module.
3. **Preheater / Heater:** Raises the feed temperature to enhance permeation.
4. **Membrane Module:** Contains the dense membrane where separation occurs.
5. **Vacuum Vessel:** Maintains low pressure on the permeate side.
6. **Condenser:** Condenses the permeate vapour back to liquid.
7. **Permeate Tank:** Collects the condensed permeate.
8. **Vacuum Pump:** Continuously removes non-condensable gases to maintain vacuum.
9. **Product Tank:** Collects the retentate (non-permeating components).

How the Driving Force is Achieved

Membrane Type

- **Dense Membranes** are used for pervaporation (similar to reverse osmosis).
 - No pores—transport occurs by **dissolution and diffusion** through the polymer matrix.
 - This is fundamentally different from porous membranes used in MF/UF.

Step 1: Sorption (Dissolution)

Components from the feed liquid **dissolve** into the membrane at the feed-side interface. The amount dissolved depends on:

- **Activity (composition)** of the component in the feed.
- **Solubility** of the component in the membrane material. Equilibrium is assumed at the interface.

Step 2: Diffusion

- Dissolved components **diffuse** through the bulk of the membrane.
- Diffusion is driven by the **concentration gradient** across the membrane.

Step 3: Desorption / Evaporation

At the permeate-side interface, components **evaporate** due to the low pressure.

The composition at this interface depends on:

- **Partial pressure** of the component.
- **Saturation vapour pressure** of the component (which depends on temperature).

Main Commercial Application: High-Purity Alcohol

Challenge:

- Ethanol and water form an **azeotrope** at 95.6% ethanol (by mass).
- Traditional distillation cannot exceed this purity.
- To produce **anhydrous ethanol** (>99.5% purity) for fuel or industrial use, additional steps are required.

Hybrid Solution: Distillation + Pervaporation

- **Distillation:** Concentrates ethanol to the azeotropic composition (~95.6%).
- **Pervaporation:** Breaks the azeotrope and removes the remaining water, producing **high-purity ethanol** (>99.5%).

Why Hybrid?

- Distillation is efficient for bulk separation (up to the azeotrope).
- Pervaporation is efficient for the final "polishing" step (breaking the azeotrope).
- **Combined:** Lower overall energy consumption than azeotropic or extractive distillation.

Membrane Used for Alcohol Dehydration

Type: Cellulose-acetate-based composite membrane.

Active Layer: Polyvinyl alcohol (PVA).

- PVA is **hydrophilic** (water-loving).
- It preferentially permeates **water** over ethanol.

Why Water-Selective?

- In ethanol dehydration, the goal is to **remove water** from ethanol.
- A water-selective membrane allows water to permeate, leaving pure ethanol as the retentate.

Other Potential Applications

Azeotropic Mixtures:

- Isopropanol-water
- Tetrahydrofuran-water
- Acetonitrile-water

Close-Boiling Mixtures:

- Benzene-cyclohexane
- Xylene isomers
- Toluene-methanol

Special Potential:

- As a **polishing step** following conventional separations.
- When **high product purities** are required (>99%).
- For **heat-sensitive** products that would degrade at distillation temperatures.

Advantages of Pervaporation

- **Breaks Azeotropes:** Can separate mixtures that are impossible by simple distillation.
- **Low Temperature Operation:** Suitable for heat-sensitive materials.
- **Energy Efficient:** For azeotrope breaking, can require less energy than azeotropic distillation.
- **Environmentally Friendly:** No entrainers or solvents required.
- **Modular and Scalable:** Membrane modules can be added or removed to meet production needs.

Case Study 1: Fuel Ethanol Production (Brazil)

Brazil is one of the world's largest producers of fuel ethanol from sugarcane. The fermentation broth contains about 8-10% ethanol. Distillation concentrates this to about 95.6% ethanol (the azeotrope). However, for blending with gasoline, anhydrous ethanol (>99.5%) is required.

Challenge: Breaking the ethanol-water azeotrope. Traditional methods in Brazil used azeotropic distillation with cyclohexane as the entrainer. This requires an additional distillation column and the recovery of the entrainer, which adds energy and cost.

Pervaporation Solution: A major ethanol producer installed a pervaporation plant using a water-selective PVA composite membrane. The azeotropic ethanol (95.6%) from the distillation column is heated to 80-90°C and fed to the pervaporation modules. A vacuum is applied on the permeate side.

Results:

- **Permeate (water-rich):** Contains 20-40% water, 60-80% ethanol. This is recycled back to the distillation column.
- **Retentate:** Anhydrous ethanol (>99.5% purity).
- **Flux:** 1.0-1.5 kg/m²·h.
- **Energy savings:** The pervaporation step consumes about 30% less energy than the azeotropic distillation it replaced.
- **Cost savings:** Elimination of the entrainer (cyclohexane) and the associated recovery system.
- **Capacity:** 100,000 litres per day of anhydrous ethanol.

Economic Impact:

- Capital investment: \$5 million for the pervaporation system.
- Annual energy savings: \$1.5 million.
- Annual entrainer savings: \$0.5 million.
- Payback period: 2.5 years.

Case Study 2: Chemical Industry - Benzene/Cyclohexane Separation

A chemical plant produces cyclohexane by the hydrogenation of benzene. The reaction does not go to completion, so the product stream contains a mixture of benzene and cyclohexane. These two components are difficult to separate by distillation due to their close boiling points (0.6°C difference) and the shallow azeotrope.

Challenge: The plant currently uses extractive distillation with N-methylpyrrolidone (NMP) as the solvent. The NMP is expensive, and its recovery requires additional energy. The plant wants to reduce operating costs and eliminate the use of organic solvents.

Pervaporation Solution: The plant installed a pilot-scale pervaporation unit using a CA-PPN composite membrane (cellulose acetate with polyphosphonate). The feed is a 50/50 benzene/cyclohexane mixture. The membrane is benzene-selective.

Results from Pilot Trials:

- **Feed:** 50% benzene, 50% cyclohexane (by mass).
- **Permeate:** 85-90% benzene, 10-15% cyclohexane.
- **Retentate:** 20-25% benzene, 75-80% cyclohexane.
- **Flux:** 0.3-0.5 kg/m²·h at 74°C.
- **Selectivity:** Benzene/cyclohexane separation factor of 8-10.

Process Design for Full Scale:

The plant decides to use pervaporation as a **polishing step** after an initial distillation. The distillation column produces a stream that is 70% benzene/30% cyclohexane. This stream is fed to the pervaporation unit. The retentate from pervaporation is 95% cyclohexane, which is the product. The permeate, which is benzene-rich, is recycled to the front of the distillation column.

Economic Impact:

- The extractive distillation column can be decommissioned.
- Solvent cost (NMP) saved: \$200,000/year.
- Energy savings: The pervaporation step requires less energy than extractive distillation.
- Payback period: 3 years.
- Environmental benefit: No solvent waste, reducing waste treatment costs.

General Membrane Equation

General Membrane Equation is used to predict and improve the performance of a membrane separation process

Complexity of Real Systems:

- Membranes have complex, non-uniform pore structures (e.g., tortuous paths).
- Solutes (molecules, colloids, particles) interact with the membrane and each other in complex ways.
- The composition of process streams (e.g., whey, juice, wastewater) is variable and complex.

Analogous Situation:

- This is similar to how we handle conventional filtration.
- We rarely predict a filter cake's exact permeability from first principles; we use empirical or semi-empirical models (e.g., Darcy's Law, Carman-Kozeny).

Consequence:

- It is currently impossible to predict flux and rejection *from first principles* for real industrial systems.
- We must rely on experimental data and a robust phenomenological model.

General Membrane Equation

Objective: To provide a framework for understanding the factors that determine the membrane permeation rate (Flux).

The Model: The equation is empirical representation based on the concept of resistances in series.

$$J = \frac{|\Delta P| - |\Delta \Pi|}{(R_m + R_c)\mu}$$

- J = Membrane Flux (Volumetric flow rate per unit area) [e.g., m³/m²s or LMH]
- $|\Delta P|$ = Transmembrane Pressure (TMP) [e.g., Pa or bar]
- $|\Delta \Pi|$ = Osmotic Pressure Difference across the membrane [Pa or bar]
- μ = Viscosity of the solvent (permeate) [Pa·s]
- R_m = Membrane Resistance [m⁻¹]
- R_c = Resistance due to fouling/ cake layer [m⁻¹]

General Membrane Equation

The Driving Force

- **Net Driving Force** ($|\Delta P| - |\Delta \Pi|$):
 - The applied pressure pushes the solvent through the membrane.
 - The osmotic pressure opposes this flow. It's the "suction" created by the higher concentration of retained solute at the membrane surface.
- **The "Threshold" Condition:**
 - For flow to occur, the TMP **must exceed** the osmotic pressure: $|\Delta P| > |\Delta \Pi|$.
 - If $|\Delta \Pi|$ is 20 bar, an applied pressure of 15 bar will result in zero flux (or even reverse flow).
- **The Osmotic Pressure Term:**
 - $\Delta \Pi = \Pi(\text{feed side}) - \Pi(\text{permeate side})$
 - Typically, permeate has negligible osmotic pressure, so $\Delta \Pi \approx \Pi_{\text{feed}}$
- **Mistake to Avoid:** The pressure you read on the gauge is $|\Delta P|$, but the *effective* pressure is $|\Delta P| - |\Delta \Pi|$.

General Membrane Equation

The Resistances

Total Resistance = $R_m + R_c$

Membrane Resistance (R_m):

- An intrinsic property of the clean, unused membrane.
- Depends on membrane thickness, porosity, tortuosity, and pore size.
- **Analogy:** Darcy's law for flow through a porous medium.
- Pure water flux is measured to determine R_m : $J = \frac{|\Delta P|}{R_m \mu}$.

Cake/Incremental Resistance (R_c): The "extra" resistance that develops during operation.

Sources of R_c :

- **Filter Cake:** Deposition of particles or colloids.
- **Gel Layer:** Macromolecules (e.g., proteins) precipitate at the membrane surface when their concentration exceeds solubility.
- **Adsorption:** Solutes adsorb onto the membrane surface or within its pores, narrowing the channels.
- **Pore Blocking:** Particles plug the membrane pores.